



Mouse Tracking for Reading (MoTR): A new naturalistic incremental processing measurement tool

Ethan Gotlieb Wilcox^{a,*}, Cui Ding^b, Mrinmaya Sachan^a, Lena Ann Jäger^{b,c}

^a ETH Zürich, Department of Computer Science, Switzerland

^b University of Zürich, Department of Computational Linguistics, Switzerland

^c University of Potsdam, Department of Computer Science, Germany

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ABSTRACT

We introduce Mouse Tracking for Reading (MoTR) a new incremental processing measurement tool that can be used to collect word-by-word reading times. In a MoTR trial, participants are presented with text, which is blurred, except for a small region around the tip of the mouse. Participants must move the mouse to reveal and read the text. Mouse movement is recorded, and, using a postprocessing pipeline we present, can be analyzed to produce scanpaths as well as word-by-word reading times. We validate MoTR in two suites of experiments. In the first experiment, we collect data for the English-language Provo Corpus (Luke and Christianson, 2018). We analyze scanpaths and show that participants interpolate between two types of strategies for reading during a MoTR trial – sometimes they fixate on individual words, somewhat akin to eye-tracking, while other times they produce a more constant pass over the text, slowing down in response to processing difficulties. Taking these strategies into account, we show that the word-by-word reading times produced by our data analysis pipeline correlate well with previously collected eye-tracking data for this corpus, and that these correlations are higher than those produced by SPR data, which we also collect for the corpus. Furthermore, we demonstrate that there is a linear relationship between by-word MoTR values and word-level surprisal values, as has been previously shown for eye-tracking data (Smith and Levy, 2013). In the second experiment, we assess whether MoTR can be used to study sentence processing phenomena in targeted psycholinguistics experiments. Using materials from Witzel et al. (2012), we show that MoTR can reveal English speakers' preferences for low attachment during online sentence comprehension. We argue that MoTR presents a compelling tradeoff between multiple experimental considerations: It is cheap to run and can be presented in a browser enabling the collection of data over the internet. It is more naturalistic than some alternative processing measures, allowing participants to skip words and regress to previous sentence regions. Finally, it has good sensitivity, detecting signatures of psycholinguistic processing behaviors from a relatively small number of participants.

Introduction

Language comprehension proceeds incrementally and non-uniformly. By INCREMENTALLY, we mean that the language processor attempts to integrate information as soon as it is encountered. By NON-UNIFORMLY, we mean that – at many levels of analysis, such as the word – language comprehension involves different amounts of effort at different times. For example, in fMRI experiments, the amount of oxygen used by the brain during language comprehension varies from word to word. And in eye-tracking experiments, the amount of time participants spend attending to an individual word varies as well. Note, however, that at other levels of analysis language comprehension and production have argued to be uniform (Fenk & Fenk, 1980; Jaeger & Levy, 2006; Meister et al., 2021); for example, evidence suggests that

people may produce language at a bit-per-second rate that is constant across the world's languages (Coupé et al., 2019; Pellegrino et al., 2011). Regardless, any descriptively adequate psycholinguistic theory must be able to predict the non-uniformity of effort observed during incremental language processing. Indeed, testing theories' ability to do so in both naturalistic and controlled settings has been a major component of theory comparison in the literature to date.

One of the major empirical resources the field has developed to test such theoretical predictions is self-paced incremental processing studies. These studies fall into at least two modalities – self-paced listening (Ferreira et al., 1996; Papadopoulou et al., 2013) and self-paced reading (Just et al., 1982), which is the focus of the present article. In the self-paced reading paradigm, participants are given a text

* Corresponding author.

E-mail address: ethan.wilcox@inf.ethz.ch (E.G. Wilcox).

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to read at their own speed and the amount of time they spend attending to each word is recorded, producing a pattern of READING TIMES (RTs). In some measurement paradigms, RTs stand for “Reaction Times;” here we will use “RTs” to refer to the measurement produced by any self-paced incremental processing study. The crucial feature of self-paced incremental processing studies is that they measure effort through time. We want to note that this idealization – more processing difficulty equals more processing time – may not always hold. For example, very short fixations are sometimes observed in very difficult processing situations, such as in gardenpath sentences. And attentional resources may not be solely devoted to the processing of the word which is the focus of a participant’s gaze. However, the effort–time link remains a useful assumption, and sets self-paced incremental processing measurements apart from other psycholinguistic techniques, which measure effort through other proxies. For example, in fMRI and EEG the amount of time for which a word is visible to the participant is often fixed by the researcher, and the dependent variables are various physiological indicators, although this is not always the case (see, e.g., Dimigen et al., 2011; Kretschmar et al., 2009; Metzner et al., 2015, 2017). Some types of self-paced incremental processing experiments can also provide data about participants’ gaze location, which can serve as a proxy for where the participant is devoting attentional resources.

Self-paced incremental processing measurements have been hugely influential in psycholinguistic theory building and theory comparison. Because of this, a variety of different methods have been developed to measure reading times, of which we will focus on three: EYE-TRACKING (Rayner, 1998; Yarbus & Yarbus, 1967), where the angle of the gaze is measured to determine fixation location during reading; SELF-PACED READING (SPR) (Just et al., 1982) where a participant presses a button to reveal and read words of a text; and MAZE (Boyce et al., 2020; Forster et al., 2009), which is an incremental judgment task, where a participant navigates through a sentence by selecting the correct continuation over a distractor, which is either ungrammatical or nonsensical.

Each of these methods has its strengths and weaknesses, presenting a tradeoff between precision, naturalness, and cost. Precise methods are good at localizing reading time slowdowns, either spatially or temporally, as well as measuring the magnitude of the slowdown. Natural methods possess a high degree of ecological validity; they allow participants to read in a way that is not contrived, and thus enable findings to generalize beyond laboratory conditions. Finally, cost can be measured both in terms of the monetary cost of the instruments themselves and also in terms of the human labor and time required to set up and run an experiment. Some methods have high precision but are expensive and laborious. Of the three methods discussed here, eye-tracking falls into this category – it has higher precision but is the most costly. Other methods, like SPR or maze, are cheaper to deploy and can be made available over the web, however, they suffer from lower naturalness and sometimes also from lower sensitivity, as is the case with SPR, which is particularly susceptible to spillover effects.

In this work, we propose a new method for incremental processing measurement which is designed to balance the higher precision and naturalness of eye-tracking with the cost and web deployability of SPR and maze. We call this method MOUSE TRACKING FOR READING, or MoTR. In a MoTR study, participants are presented with a text that is blurred, except for a small region around the tip of the mouse, which is in focus. The design of the blurred and focused regions is intended to loosely mimic the human visual field, with its division between foveal and parafoveal vision. The participant is instructed to move the mouse to reveal and read the text. Participant mouse location is recorded and analyzed to produce a scanpath, as well as the amount of time spent attending to individual words. An example of a MoTR trial is given in Fig. 1, and a live demo of a MoTR experiment can be found at the following link: https://wilcoxeg.github.io/MoTR/MoTR/run_motr_in_magpie/demo/. Although the idea of using occlusion and highlighting to study human attention in an experimental setting has

been recently proposed by Anwyl-Irvine et al. (2021), we are the first to adapt this method for studying language processing. The entire MoTR paradigm can be run in a browser, making it cheap to deploy. At the same time, it allows for more naturalistic reading because in MoTR participants can skip words, and regress to previous portions of the sentence, something that is not available generally in maze or SPR.

We argue that MoTR can be an attractive alternative for eye-tracking and other processing measurement tools, especially as a way to increase the diversity and representativeness of psycholinguistic data. Psycholinguistic theories are intended to describe human language *in general* and are thus expected to generalize across languages. However, much of the data that has driven the development of psychological theories is not particularly diverse, either in terms of the languages it represents or the types of populations from which it is collected. Although we validate MoTR using English data, we believe that it can be used effectively for languages where no laboratory infrastructure is available, or in the field, where it requires only a computer with a mouse or track-pad. We hope that more naturalistic yet lower cost methods such as MoTR will contribute towards a better crosslinguistic picture of sentence processing, which, in turn, will help to increase the generalizability of psycholinguistic theories.

In the rest of this paper, we describe the MoTR paradigm and validate it in two suites of experiments. In the first experiment, we collect MoTR data for a corpus of written English texts (the Provo Corpus; Luke & Christianson, 2018) and conduct three analyses: First, we inspect the scanpaths produced by participants to provide an overall sense of how they are using the tool. Second, we compare the word-by-word reading times produced by MoTR to eye-tracking data, which was collected for the initial release of the corpus, as well as with self-paced reading data which we collect ourselves. In the second suite of experiments we assess MoTR’s ability to replicate, in controlled experiments, well-established findings that have been important for psycholinguistic theory building. Using MoTR, we collect data for native English speakers’ processing of high vs. low attachment on a suite of materials that have been used previously to compare maze, SPR, and eye-tracking (Witzel et al., 2012) as well as maze and SPR (Boyce et al., 2020).

Our results indicate that MoTR can be a viable and attractive alternative to other self-paced incremental processing methods. While using the tool, we find that participants often interpolate between two strategies: First, they will sometimes produce fixations and rapid mouse movements, somewhat similar to eye-tracking. However, they will also often produce a smoother pass over the text, slowing down during words that are difficult to process. Our data-processing pipeline, which turns these scanpaths into word-by-word reading times, measures the amount of time a participant’s mouse is closest to each word and is thus compatible with both types of strategies. For the Provo dataset, we find that the word-level MoTR RTs produced by this pipeline are well correlated to the eye-tracking and self-paced reading data, as well as to word-level statistical properties such as a word’s log frequency and length. In a replication of Smith and Levy (2013) on the Provo data, we find that the functional relationship between MoTR reading time and a word’s surprisal – which measures its informativity – is roughly linear, a result that has been previously established for eye-tracking data. In our targeted experiment, we find that MoTR RTs replicate preferences, for the given materials, for low attachment of prepositional phrases and relative clauses and produce magnitudes between conditions that are similar to eye-tracking. Further, we find that MoTR data provides novel information about the role of regressions when processing different attachment ambiguities, something that has not been reported previously in the literature.

In sum, we are optimistic about the ability of MoTR to offer a cheaper alternative to eye-tracking that can capture many of the same language processing effects. Although we validate our new paradigm on native English speakers, we hope that MoTR can enable data collection from a more diverse sample set of participants. We release code to run

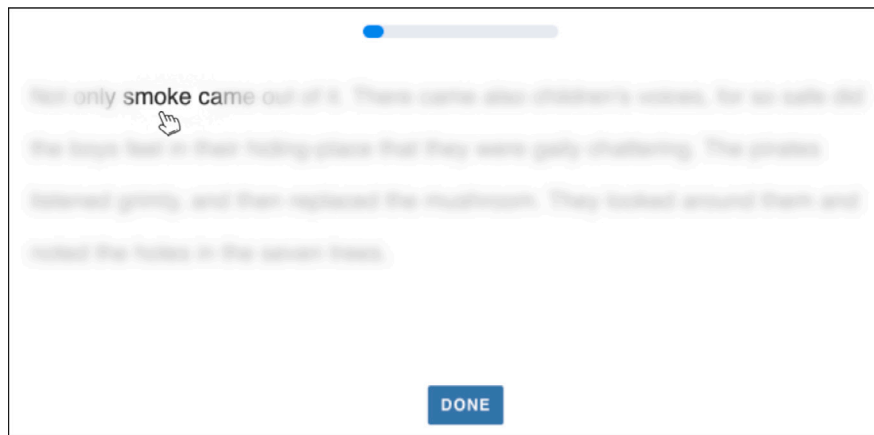


Fig. 1. A MoTR trial: The participant moves their mouse to reveal and read the text.

MoTR experiments and process MoTR data along with the publication of this article.¹

Background: Incremental processing measures

Eye-tracking-while-reading

Eye-tracking records where and when participants direct their gaze as they freely read texts displayed on a computer monitor. When properly collected, it is highly precise, both at the spatial and temporal level, and is considered the gold standard for incremental language processing research (Holmqvist et al., 2011).² Here, we discuss video-based eye-tracking. While other eye-tracking methods exist, video-based devices that use the pupil-and-corneal reflection method are typically used in psycholinguistics research. As suggested by its name, in the pupil-and-corneal method two elements on the eyeball are targeted: the pupil and the cornea. The pupil is the hole in the center of the eye that admits light; the cornea is a transparent film that covers the surface of the eye. Its reflection of light produces a glint, which is called the 1st PURKINJE REFLECTION (P1). Eye trackers work by using infrared light and cameras to locate the P1 which, when compared to the location of the pupil, enables the determination of the eye's orientation and gaze direction. By calibrating these with the eye's geometry and monitor distance, the precise gaze location on the screen is computed (Holmqvist et al., 2011). Apart from its high temporal and spatial resolution, eye-tracking has several advantages over other methods. First, it allows unconstrained movement of the gaze through the text, meaning that eye-tracking can measure both forward and backward saccades. This is necessary, for example, to study things like regressive eye movements or word skipping. Second, eye-tracking is natural insofar as it resembles the type of reading participants are likely to do in non-experimental settings. Little or no training is required to get participants to produce the required experimental behavior.

Some recent studies have taken advantage of consumer electronic devices that come equipped with cameras to run eye-tracking experiments on larger and more diverse samples of participants including some web-based experimental paradigms (Papoutsaki et al., 2016; Semmelmann & Weigelt, 2018; Xu et al., 2015). For example, Degen et al.

(2021) and Vos et al. (2022) replicate the visual world paradigm using laptop cameras. And Werchan et al. (2022) have used webcams to record gaze direction in infant populations. Recently, Ribeiro et al. (2023) used web cameras to conduct eye-tracking while reading studies. While they find some correlations between word-by-word reading times collected from laptop cameras and recordings from laboratory grade eye-tracking, these are relatively low (i.e., a Spearman rank correlation of ≈ 0.55), and they vary greatly based on the quality of a participant's web camera. While these studies are undoubtedly useful for some paradigms, such as the visual world paradigm, the low resolution and difficulty of calibration for laptop cameras suggest that they are infeasible for eye-tracking while reading studies.

Another recent method, *MouseView.js*, uses mouse tracking plus blur and spotlight as we propose here (Anwyl-Irvine et al., 2021). When compared against eye-tracking data for scanpaths of images, the tool produces similar dwell times as in-lab eye-tracking. MoTR and *MouseView.js* are, essentially, the same technique, however MoTR is optimized to collect data for reading studies.

Self-paced reading (SPR)

Moving window SPR

Self-paced reading comes in several variants. The most prevalent one is the moving-window SPR, where participants press a button to reveal each subsequent word of a text while simultaneously masking the previous word. The time taken to press the button is recorded and used as a proxy for incremental processing difficulty. Although undoubtedly more constrained, it is less costly. To run an SPR experiment in the lab, one needs only a standard personal computer without any specialized hardware, which is much cheaper than an eye-tracker. In addition, the relative ease of an SPR experiment means that experimenters can handle multiple participants in parallel (in contrast to eye-tracking, which requires a one-to-one setup with one experimenter running one participant at a time). Finally, recent web-based experimental software has made it easy to run SPR experiments in a web browser, where a single researcher can launch an SPR experiment that is taken by dozens of participants remotely.

At the same time, SPR faces several drawbacks primarily because it is less naturalistic than eye-tracking. Unlike in typical reading, in traditional moving-window SPR participants do not have the option to regress to previous words, and participants cannot skip words directly. There are variants of SPR that do allow regressions, including the CUMULATIVE WINDOW SPR, where words remain on screen after their initial presentation. While cumulative window SPR may help with the measurement's naturalness, it does not enable the researcher to record participants' regressions. Furthermore, several studies have reported that in cumulative window SPR participants will often press the 'next'

¹ Data and code for both MoTR and SPR data be found online at <https://osf.io/4g7pr>.

² For spatial precision, one typical measure is the root mean square (RMS) of the inter-sample angular distances during a fixation. Typically, for a reading experiment, an RMS of $< 0.05^\circ$ is considered sufficient. For temporal precision, eye-trackers can sample at a rate of up to 2000Hz; For reading research, a sampling rate of 500Hz is considered sufficient (Poole & Ball, 2006; Rayner et al., 2012).

button in rapid succession, and only commence reading after the entire text has been revealed, meaning that the experimental data is decoupled from processing times (Ferreira & Henderson, 1990; Just et al., 1982). One other drawback of SPR is that it yields limited temporal resolution because it is susceptible to *SPILLOVER EFFECTS*, where the slowdown in response time caused by a processing difficulty on one word carries over and manifests on subsequent words (Boyce et al., 2020; Smith & Levy, 2013; Witzel et al., 2012). Spillover effects are also present in eye-tracking, although to a lesser extent. Recently, data analysis techniques have been developed to effectively deconvolute the SPR signal (Shain & Schuler, 2021). One area where SPR may be advantageous over eye-tracking is in the case of *PREVIEW EFFECTS*. During eye-tracking, participants are sometimes aware of words ahead of their current fixation point due to parafoveal vision. This is not possible in SPR due to word masking, and therefore SPR can be used in cases where researchers wish to control or eliminate the effect of parafoveal preview. Of course, whether or not this feature of SPR presents an advantage or disadvantage depends on one's research question. For example, because it precludes preview effects, SPR may not be an appropriate method for studying parafoveal processing, including, for example, parafoveal-on-foveal effects.

Bidirectional SPR

Recently, Paape and colleagues (Paape & Vasishth, 2021; Paape et al., 2022b) proposed Bidirectional SPR (BSPR), a variant of SPR that allows for regressive movement. A BSPR trial is similar to an SPR trial, except that instead of only moving forward through a sentence participants can press a button to reveal the previous word or sentence region as well. Regressions in BSPR are associated with lower acceptability scores (Paape & Vasishth, 2021) and in a post-experimental questionnaire, participants report that BSPR feels more “natural” than standard SPR. However, as the method requires participants to click through previous words or sentence regions as opposed to making a saccade, regressions were found to be relatively rare in BSPR trials, leading the authors of the paper to suggest that they are used as a last resort strategy by participants. Nevertheless, BSPR shows promise as a more natural extension of traditional SPR methods. Because BSPR is the only web-deployable incremental processing measurement tool that can record the location of regressions more precisely, in Section “Experiment 1: Naturalistic Reading in English” we present a side-by-side comparison of BSPR and MoTR.

Maze

In the maze task (Forster et al., 2009) a sentence is displayed word-by-word in discrete trials. For each trial, both a correct word and a distractor word are displayed on the screen. Participants are required to select the correct word which allows for a plausible continuation of the sentence. The duration of the selection process is taken as a proxy for incremental processing difficulties. For example, given the sentential history, *The elephant stepped on the ...* participants might be presented with *ant* and *and*, with the former being the correct continuation and the latter being the distractor.

Maze comes in two variants: the Grammatical maze task (G-maze) and the Lexical maze task (L-maze). In the G-maze, the distractors are real words, but they do not form a grammatically correct continuation of the sentence. In contrast, the L-maze employs nonce words as distractors. Recently, Vani et al. (2021) proposed a novel variant, I-maze, which interleaves L-maze and G-maze trials. Like SPR, maze is deployable on a laptop or over the web, meaning that it can be run cheaply. In addition, because it forces strongly incremental processing, maze data does not suffer from the spillover effects observed in SPR (Boyce et al., 2020, 2023; Witzel et al., 2012). Finally, because maze is strongly incremental, it is not susceptible to preview effects.

One problem with the original formulation of the maze task is that distractor selection for G-maze can be difficult and time-consuming.

Distractors and targets must be matched for a range of lexical properties such as orthographic length and frequency, and distractors must constitute an unacceptable sentence continuation. To alleviate this burden, Boyce et al. (2020) propose A-maze, a technique for automatically generating distractor words using contemporary neural-network-based language models. They compare both in-lab and web-based A-maze experiments to SPR, demonstrating that the maze paradigm typically recovers larger effect sizes between conditions for controlled psycholinguistic tests.

The major remaining drawback of the maze is that it is the least naturalistic of the different paradigms discussed here. In a maze trial participants must read not one but two words, decide which one is the target, and then press the corresponding button. Thus, while maze RTs can be analyzed in a similar way to RTs from other paradigms (i.e., as measures of incremental processing difficulty), they tend to be quite a bit longer. Additionally, maze does not allow for word skipping or regressive passes over a text.

Mouse tracking for reading (MoTR)

A MoTR trial

General description of a MoTR trial

In each trial, participants are presented with a screen that displays text, which is blurred except for a small region around the tip of the mouse. We will refer to this in-focus area as the *SPOTLIGHT*. Participants are instructed to move the mouse, bringing individual words into focus in order to reveal and read the text. A screenshot from a single trial, showing the text blur and spotlight, is given in Fig. 1. During the trial, participant mouse movement is recorded. The coordinates of the cursor are then analyzed as a proxy for gaze. In our implementation, when participants have finished reading the trial text, they press a button at the bottom of the screen, which removes the spotlight and reveals a comprehension question. Participants can proceed to the next screen only after they have answered the comprehension question. All of our experiments are implemented in Magpie (<https://magpie-ea.github.io/magpie-site>), which is a general framework for running browser-based psychological experiments.

Spotlight size and cursor appearance

The size of the spotlight needs to balance two considerations. If the spotlight were too large, say encompassing 4–5 words or spanning multiple lines of text, then it could be difficult to know which word the participant was attending to at any given moment. However, if the spotlight were too small, participants would be forced to make extremely quick, incremental mouse movements, which could be both frustrating and potentially physically difficult, depending on the sensitivity of the participant's mouse or trackpad. We ran initial pilot experiments with varying spotlight sizes, ranging from 80 pixels to 150 pixels. These data suggested that participant behavior does not qualitatively change above 100 pixels. Thus, for our experiments, we set the spotlight width to be 102 pixels. (The extra 2 pixels is done to accommodate the increased size of boldfaced font.)

One other consideration has to do with the appearance and placement of the cursor. One potential worry is that if the cursor were fully absent, then participants would have a difficult time navigating over white space to the beginning of the text, or to the *done* button at the bottom of the screen. However, if the cursor were horizontally aligned with the center of the spotlight, then it would interfere with reading. As a compromise, we place the cursor below the horizontal center line of the spotlight, and slightly vertically off-center to the left. One additional benefit of the cursor is that it may direct participant gaze during reading, although we do not test this experimentally. The off-center alignment is done to mimic the asymmetric nature of the human gaze during reading, which typically picks up 5 characters to the left of the fixation point, and 15 characters to the right (McConkie

& Rayner, 1976; Rayner et al., 1980). In our experiments, the center of the cursor is 39 pixels from the left edge of the spotlight and 63 pixels from its right edge. Note that the size of the spotlight was chosen to match our text size, and is not a core feature of the MoTR paradigm. Given that the text size and cursor spotlight size are parameters in our Magpie implementation, it is easy to change them should studies require different fonts or different font sizes.

As one final feature, the spotlight window incorporates a gradual transition effect along its edges, creating a smooth progression from focused to blurry, which mimics the central focus and diminishing clarity towards the periphery in human reading (Rayner, 1998). Fig. 1 demonstrates the cursor alignment and asymmetric cursor placement. Because of the gradual blur transition, about five letters to the left of the cursor are fully clear, while an additional four are somewhat obfuscated. On the left side about two letters are fully clear, while an additional four are partially obfuscated.

Appearance and placement of text

In our experiments, text is double-spaced on the page. This is done for two reasons: First, participants varied in terms of the vertical placement of the spotlight relative to the text line, and we did not want lower or higher placement revealing adjacent lines of text. Second, text double spacing makes our experiments a closer match to eye-tracking setups, where text is double-spaced to account for vertical calibration errors.

The purpose of the text blur is also twofold. First, it was done to (loosely) mimic human vision in the periphery, which conveys low-resolution information such as the location of line breaks (McConkie & Rayner, 1975; Rayner, 1975; Strasburger et al., 2011). Second, the blur was necessary to obfuscate enough material so that the participant must move the mouse to reveal the text. We choose to blur the text, as opposed to, say, blacking it out, because this presentation still retains some general information about the placement of the text on the screen, which may be useful for navigating to the beginning of the text or to the start of a line. For our experiments, the blur effect was implemented with the standard CSS `blur()` function with a radius of 3.5 pixels.

Sampling rate

In order to record the location of the spotlight, we continuously sample the placement of the cursor during the experiment and adjust accordingly. We used the default data handling procedure in Magpie, which streams the data continuously from the web interface to a server during the experiment and sends additional participant information, such as the participant's ID, along with each timestamp. This means that, in total, we stream about 120 B per time stamp. We experimented with sampling rates of 40 Hz and 20 Hz. As with the spotlight size, the sampling rate needed to balance two considerations. On the one hand, we wanted a relatively high sampling rate to be able to compute precise durations of reading times. On the other hand, a high sampling rate means we need to send large amounts of data to our server during MoTR trials. With our initial sampling rate of 40 Hz, which corresponds to about 288 KB per minute, we observed some time-out issues during initial pilot experiments and thus chose to decrease the sampling rate to 20 Hz. Overall, we found that the sampling rate was stable in our experiments. For example, for data collected on the Provo corpus, the mean time between samples was 50.1 ms, with a standard deviation of 7.1 ms.

After our experiments had been completed, we identified all redundant information that was being streamed by the default protocol in Magpie, and introduced changes into the Magpie backend which eliminated these redundancies. This new protocol streams about 68 B of data per time stamp, or about 81 KB per minute, which we anticipate will allow for even higher sampling rates.

Post processing

Attentional association identification

In eye-tracking, post-processing pipelines are built around identifying fixations, or periods between saccades when the gaze is stable and focused on an individual point. However, in MoTR we cannot be sure that participants use the mouse to produce analogous fixation-plus-saccade behavior. It may be the case that participants produce more of a smooth scan over the text, slowing down or speeding up depending on the level of current processing difficulty. (For a more detailed analysis of participants' scan paths in MoTR see Section "Analysis 1: How do participants use MoTR?"). Therefore, instead of identifying fixations, our data processing pipeline is based around what we call ATTENTIONAL ASSOCIATIONS (or just ASSOCIATIONS for short). We compute associations in the following manner: The raw data from our web application produces a list of time-stamped samples of x/y screen coordinates indicating the location of the spotlight (the specific location being recorded is the slightly off-center location described above). We associate each spotlight location with a word on the screen, or else values indicating that the participant is hovering between text rows or off the edge of the window. In order to turn this data into the attentional association, we merge all consecutive samples that occur over the same word, giving the amount of time that word was at the center of the spotlight and thus most prominent to the participant. These associations are then turned into word-by-word reading measures, as described below.

There are two further details to note about this part of the pipeline: The first has to do with cases where participants' spotlight was not over the trial text. Participants were determined to be hovering *between* text lines if the upper edge of the spotlight lay below the center of the line, meaning that the top half of each character is not displayed acutely. The spotlight was determined to be off the edge of the screen *horizontally* if its right edge did not overlap with the first character of the line, or if its left edge did not overlap with the last character of the line. It was determined to be off the edge *vertically* if its upper edge lay below the center of the last line, or if its lower edge was above the center of the first line (i.e., the same criteria to determine if it was between rows). The second detail is that, for attentional association identification, we included a maximum value, f_{max} , and minimum value, f_{min} , and filtered out associations that were either too long or too short. For f_{min} we wanted to make sure that times when participants were scanning over the text very quickly were not counted as a series of very short associations, which could result from an f_{min} of zero. For f_{max} , we believe it is useful to identify and remove a handful of outlier cases that result from web-based data collection. We intend both values to be parameters that can be set by the experimenter rather than hard-coded features of a MoTR experiment. Looking at the raw data, we observed that MoTR associations tended to be longer than eye-tracking fixations. As eye-tracking minimum fixation thresholds have been typically set to 80 ms (Luke & Christianson, 2018; Prasse et al., 2020; Slattery & Parker, 2019), we explored a range of values from between 1.5x–3x this range (i.e., 120 ms–240 ms). We did not observe much difference in the data, so for the experiments we simply set f_{min} to 160 ms. While in eye-tracking studies the maximum threshold is typically set to around 1200 ms (Luke & Christianson, 2018; Luke & Henderson, 2016; Prasse et al., 2020; Slattery & Parker, 2019), we opt for a much higher upper boundary of 4000 ms.

Computing reading time measures

From the pattern of associations, our post-processing script computes the following mouse-based metrics, which are analogous to well-used gaze-based metrics from the eye-tracking literature: GAZE DURATION, which is the length of the first association on the word, before the reader either moves forward or regresses backward; TOTAL DURATION, which is the sum of the durations of all associations on a word; GO PAST TIME which is the sum of all associations durations starting from the first first-pass associations on a word until associations on

a word to the right of this word. This is also referred to as the inclusive regression-path duration. Following standard practice in the eye-tracking literature, skipped words are not counted in computing duration measures. Following Jakobi et al. (2024), Luke and Christianson (2018), we also extract the following binary measures: FIRST-PASS ASSOCIATION (FPASC) is 1 if the word was associated to in the first-pass, otherwise 0; FIRST-PASS REGRESSION (FPRREG) is 1 if a regression was initiated in the first-pass reading of the word, otherwise 0. From these binary measures we calculate cross-participant averages to produce first-pass regression probabilities and first-pass association probabilities for each word. If a reader does not associate on the word during their first pass, then this is not counted towards the first pass regression probability.

Experiment 1: Naturalistic reading in English

In this experiment, we ask whether MoTR can be used to investigate naturalistic reading in English. Participants read the Provo Corpus (Luke & Christianson, 2018), which consists of short texts from various genres. We selected the Provo Corpus because it is an eye-tracking dataset that is well-used in the psycholinguistics literature and presents data from a relatively large sample of participants. To get a sense of whether MoTR can be used to investigate patterns of naturalistic reading we ask three questions: First, how are participants using MoTR during the experiment (Section “Analysis 1: How do participants use MoTR?”)? Second, do MoTR association times correlate with data from other incremental processing measurements, as well as with word-level statistical properties (Section “Analysis 2: Comparing MoTR, Bidirectional SPR, and eye-tracking”)? And finally, what is the relationship between MoTR association times and a word’s information content (Section “Analysis 3: The Functional Relationship Between MoTR and Surprisal”)?

Materials and methods

The Provo Corpus comprises fifty-five short passages of text, ranging from between 39–58 words, drawn from a variety of genres, including literary fiction, non-fiction, and newspaper articles. Eye-tracking data from 84 native English speakers has been previously collected for this corpus, meaning it is a good testing ground to explore the differences between eye-tracking and MoTR. As the Provo Corpus was not created with comprehension questions, we wrote simple yes/no comprehension questions for each separate passage. The questions were intended to verify whether participants had understood the basic meaning of the passage after reading it. Note that the presence of comprehension questions is one difference between the original eye-tracking protocol and our MoTR protocol.

The experiment was run using Magpie and hosted on GitHub Pages. Our back end was implemented using Heroku. We collected MoTR data from 101 participants. Because the corpus was relatively large, we split it into three sub-experiments, meaning each passage was read by ≈ 30 participants; subjects could not participate in more than one sub-experiment. The MoTR trials were conducted in the same way as described in Section “Mouse Tracking for Reading (MoTR)” (for example, with the same line spacing, spotlight size, and sampling rate). At the beginning of the experiment, participants were presented with an instruction screen that introduced them to the MoTR task. The instruction screen displayed the following text: *In this study, you will read short texts and answer questions about them. However, unlike in normal reading, the texts will be blurred. In order to bring the text into focus move your mouse over it. Take as much time to read the text as you need in order to understand it. When you are done reading, answer the question at the bottom and click “next” to move on.* Participants were recruited on Prolific (<https://www.prolific.co>), and had to be native speakers of English as well as have IP addresses from within the United States. Participants were paid for their time.

We used the following data-filtering steps: First, if a participant answered fewer than 80% of the comprehension questions correctly, we removed data from that participant. This step applied to MoTR data only and resulted in the removal of 9 participants. The 80% threshold was chosen because it was the one used in Boyce et al. (2020). Second, data from individual trials was excluded if associations were detected for fewer than 20% of the total words on the screen, indicating that they were simply skimming. Skimming is a type of reading strategy where readers aim to understand the gist of a text, rather than its full semantic content. It is associated with more word skipping and longer saccades (Liao et al., 2021). Finally, RTs were excluded if they were below f_{min} or above f_{max} , as described in Section “Mouse Tracking for Reading (MoTR)”. Eye-tracking data has been post-processed as described in the original paper. This includes filtering out fixation times that are less than 80 ms and greater than 800 ms, but no filtering based on comprehension questions, as the eye-tracking materials did not include comprehension questions.

In addition to MoTR data, we also collect Bidirectional Self-Paced Reading (BSPR) data for the same materials. This was done because, like MoTR, BSPR can be collected over the internet and also provides participants with the opportunity to regress. Therefore, comparing it directly to MoTR and eye-tracking data can reveal the strengths and weaknesses of each approach. To collect BSPR data, we adapted the code provided by Paape et al. (2022a), shifting from region-by-region planned sentence reading to more naturalistic word-by-word text reading. Since the original Ixex farm platform (Drummond, 2017) used by Paape et al. (2022a) is no longer supported, we conducted and hosted our experiments on the PCIBex farm (Schwarz & Zehr, 2021). Following the same participant screening for MoTR, we recruited 90 participants. BSPR reading measures were calculated using the same post-processing pipeline as used for MoTR, including f_{min} and f_{max} suggested by Paape et al. (2022a) ($f_{min} = 150$ ms; $f_{max} = 5000$ ms), meaning that participants could technically skip over a word if the time between their button presses was below f_{min} . We also used the same criteria for data filtering on the BSPR data, resulting in the exclusion of 13 participants who scored less than 80% on comprehension questions.

Analysis 1: How do participants use MoTR?

First, we want to give a general sense of how participants are using MoTR, and how use of the tool varies between participants, items, and trials. To start off, in Fig. 2 in the left panel, we visualize the x -axis mouse movement for one participant reading a portion of the Provo corpus. We chose this participant because, as we shall see later, they are a good example of the behavior of the “average” subject in this study. Based on this visualization, it is clear that MoTR is unlike eye-tracking, which consists entirely of fixations and extremely quick saccadic movements. If participants were using such a strategy, then the line in the figure would either be fully horizontal (during a fixation) or nearly vertical (during a saccade), which is not the case. However, it is also clear that this participant is not moving over the text with constant velocity. Rather, they are interleaving between two types of strategies in response to processing difficulties: Sometimes, they completely stop x -axis forward movement and produce a true fixation. For example, a true fixation occurs at around 15000 milliseconds on the word *scientifically*. However, other times the participant merely decelerates in response to a word. One good example of this is in the word *Millenials*, where the participant reduces their velocity but continues to move their mouse forward.

Based on this observation, we divide participant behavior into five categories, which are visualized in the right panel of Fig. 2. Here, each point on the line represents a single sample from the trial, occurring roughly every 50 ms, and points are colored to their movement type. The basic types are: True fixations (red), where the velocity of the mouse measured at that point is zero; decelerations (orange); constant movement (light green), accelerations (dark green); and off-stimulus

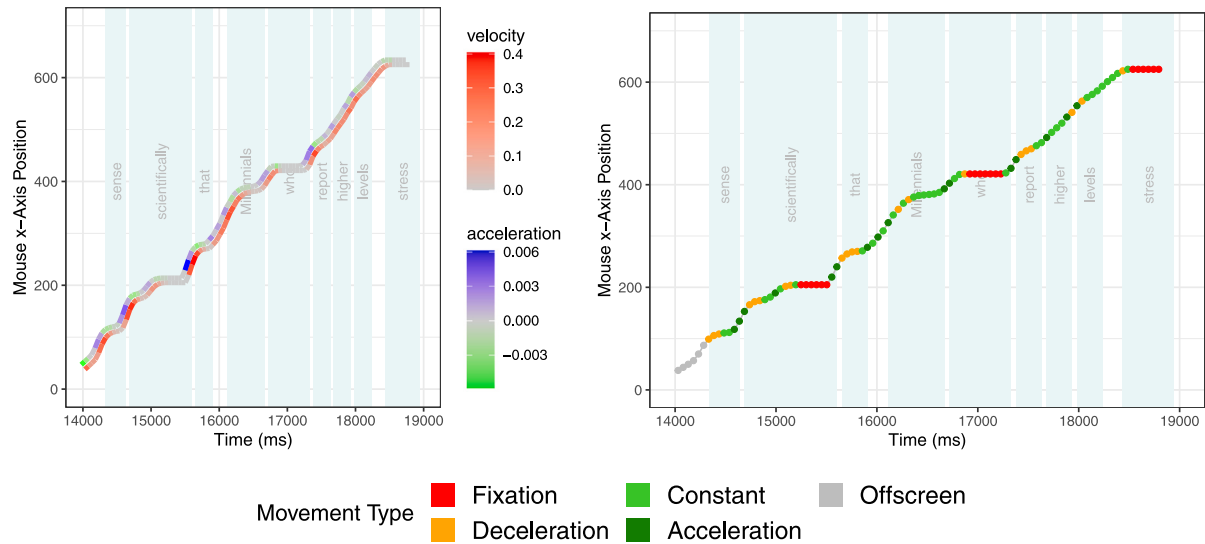


Fig. 2. MoTR reading strategies: The left panel shows the velocity (red) and acceleration (blue/green) for one participant reading a portion of the Provo corpus. The right panel shows the samples taken during this period, where each dot is a single sample. Based on their velocity, acceleration, and location, these belong to either true fixations, decelerations, accelerations, movements of constant velocity, or offscreen time. Light blue background bands indicate associations with words, which are used to compute reading measures. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

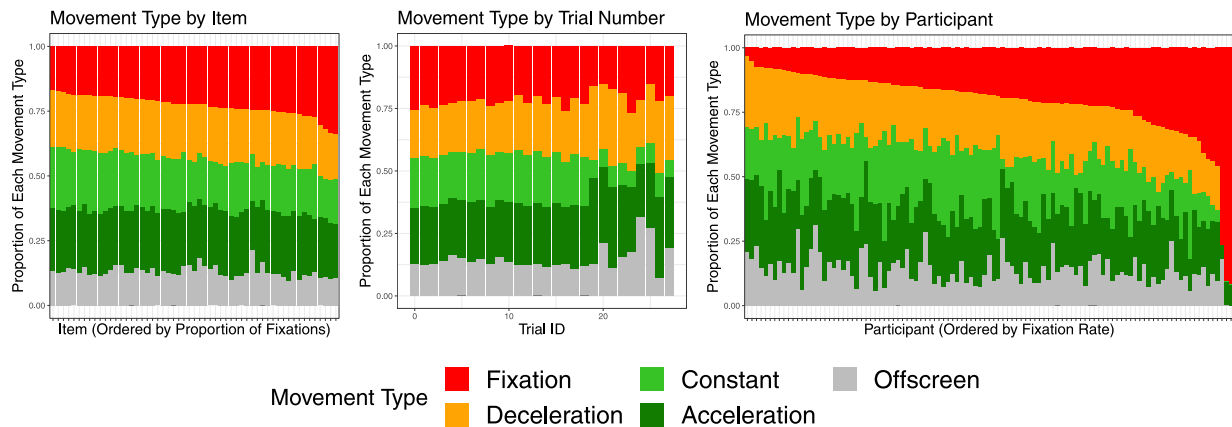


Fig. 3. Movement type by trial, item, and participant: Bars show the proportion of samples from each movement type, categorized as true fixations (red), decelerations (orange), constant velocity (light green), acceleration (dark green) and offscreen time (gray). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

time (gray), where the mouse is not revealing any text. As it may be difficult to produce true constant velocity with the mouse, we count a sample as occurring during constant velocity if the absolute value of the acceleration is under a threshold δ . For this analysis, we set $\delta = 0.001$ px/ms. One important point is that both fixations and decelerations are compatible with our approach for turning the raw data into “associations” and finally into reading measures. Because associations are based on the amount of time a cursor is closest to a given word, decelerating movements will contribute to longer association times and thus to longer reading times.

How does the use of MoTR change during an experiment, between items, and between participants? To visualize these differences, we plot the proportion of movement types broken down by various factors in Fig. 3. On the left panel, we break movement type down by the different items (i.e., short paragraphs) in the Provo corpus, ordered by the proportion of samples that were counted as true fixations. Although there is some variation, we observe roughly a similar proportion of fixations between items, ranging from 16% to 34%, with an average of 23%. Breaking down fixation by trial type (middle panel), again,

we observe relative consistency throughout the experiment. Turning to the type of movement by participant (right panel), we find more variation. In particular, we observe a handful of participants whose behavior is similar to eye-tracking, and who produce primarily fixations and saccadic-like jerky movements with the mouse. That being said, the median participant is relatively well-balanced between the five types of movements. It is from this middle distribution that we drew the participant whose movements are visualized in Fig. 2.

To give a better sense of the range of participant behavior, Fig. 4 visualizes reading from four different participants whose use of MoTR differs. In these plots, the sharp downward lines indicate times when participants move their mouse to the beginning of the next line of text. The blue bands indicate the associations, which are the result of our post-processing pipeline and contribute towards a word’s reading time measures. Orange lines show the participant’s y-axis location. The top row shows a participant who produces a smooth scan across the text, and who slows down, rather than fixating, at unexpected words, for example around *prayer* around 16000 ms. The middle two rows show participants who produce more fixations. Finally, the bottom row shows

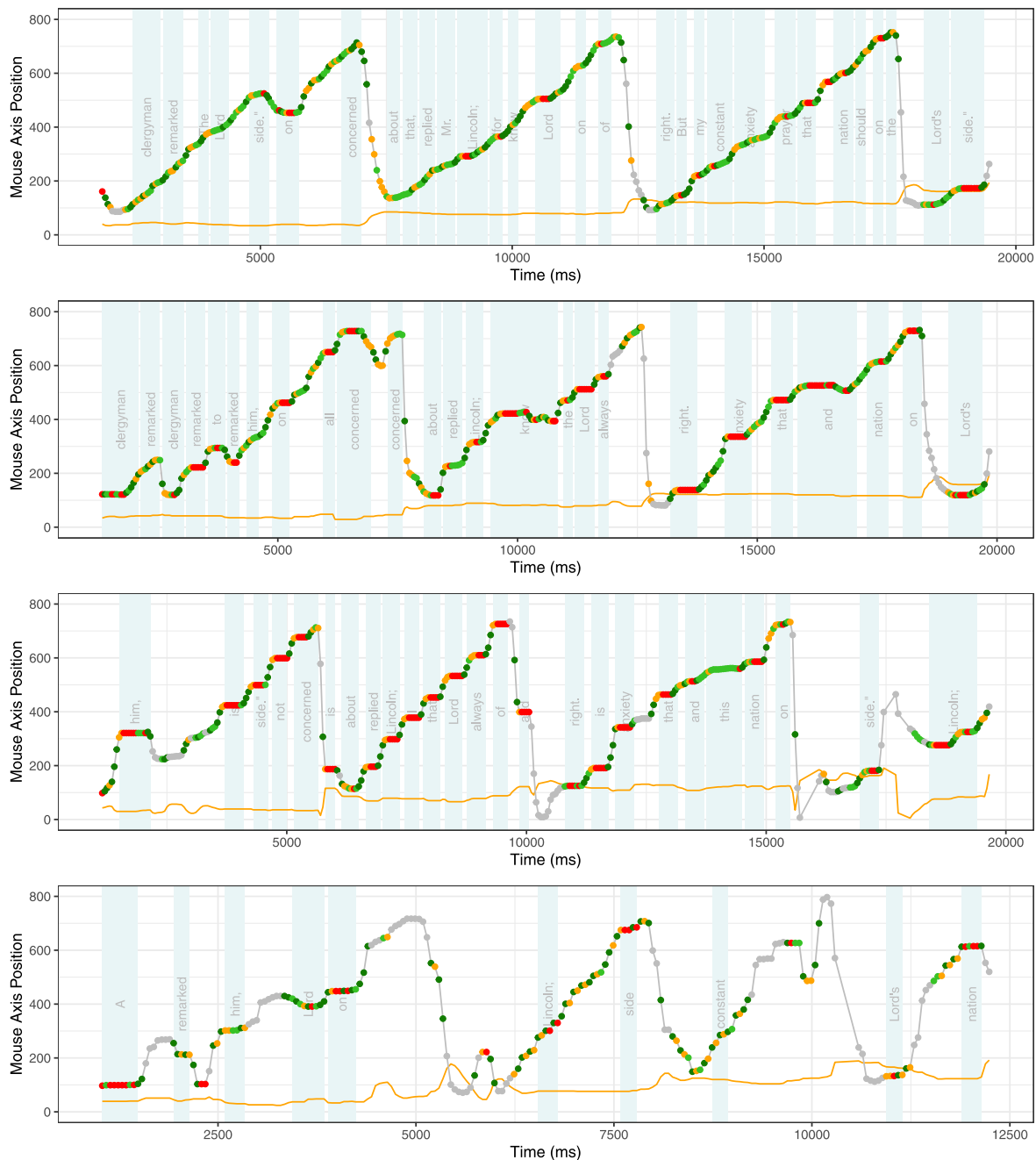


Fig. 4. Examples of different reading strategies: Smoother movements with no fixations (top row), fixation-based strategies (middle rows), and more random mouse movements (bottom row). The orange line indicates the y-axis position and is timed with sharp decreases in x-axis position, which are movements to the beginning of the next line. In this case, the bottom-row trial would be filtered, as the participant associates with fewer than 20% of total words. The text is: *A clergyman remarked to him, “The Lord is on our side.” “I am not at all concerned about that,” replied Mr. Lincoln; “for I know that the Lord is always on the side of the right. But it is my constant anxiety and prayer that I and this nation should be on the Lord’s side.”*

a participant who is using the tool *not* as intended, with lots of quick, seemingly random movements and offscreen time. In this case, this trial would be excluded during our data filtering step, as the participant associates with fewer than 20% of the total words in the text.

Analysis 2: Comparing MoTR, Bidirectional SPR, and eye-tracking

For this analysis we compare MoTR to BSPR by asking how well each measurement correlates with eye-tracking, as well as with word-level statistics. For these analyses, words that were skipped on their first pass were dropped from the dataset to calculate gaze duration and go past time, as is standard practice. Only words that were fixated

on during the first pass were included in the calculation of the first-pass regression probability. Skipped words were not included in the computation of regression probabilities because landing on a word is a precondition for the decision to regress at that word. As noted above, when looking at the relationship between MoTR and eye-tracking, all analyses were run on cross-participant averages of reading times, as was done in the prior literature, e.g., [Smith and Levy \(2013\)](#), [Wilcox et al. \(2020\)](#) (although, cf. [Hoover et al., 2023](#)).

Correlation with eye-tracking data

To compute correlations, we ran the Bayesian correlation analysis proposed in [Behseta et al. \(2009\)](#), which is more robust to outlier effects

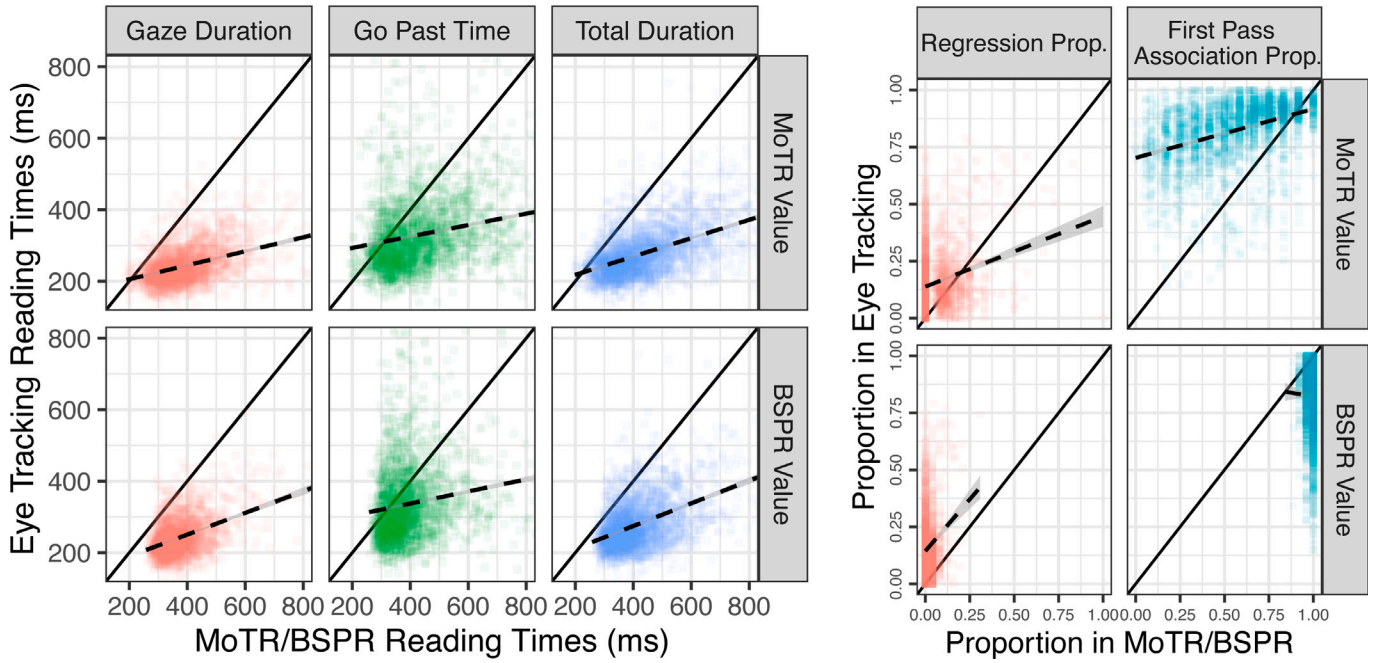


Fig. 5. Correlation between MoTR/BSPR and eye-tracking: The solid black line is the $x = y$ line. The dotted black line is the linear line of best fit.

Table 1

Correlation with eye-tracking: Posterior means and 95% Crls, centered around the mean. The top two rows show the correlation over the whole dataset. Bottom row shows the correlation between two splits of the eye-tracking dataset as an upper bound.

	Gaze duration	Go past time	Total duration	Regr. prop.	Skip prop.
MoTR	0.51 [0.48, 0.54]	0.42 [0.39, 0.45]	0.62 [0.59, 0.65]	0.19 [0.14, 0.24]	0.80 [0.78, 0.81]
BSPR	0.35 [0.31, 0.39]	0.26 [0.22, 0.30]	0.40 [0.36, 0.44]	–	–
Eye-Tr. w/ self	0.75 [0.73, 0.78]	0.63 [0.60, 0.66]	0.81 [0.79, 0.82]	0.58 [0.55, 0.61]	0.91 [0.90, 0.91]

than standard Pearson correlations (Matzke et al., 2017). For each correlation, we report the posterior means and 95% credible intervals for the correlation coefficient, r . For reading time correlations, we assume a bivariate t distribution as the likelihood, which can be seen as a generalization of the bivariate normal distribution assumed by Pearson's correlation test. For regression probability, models achieved poor fit assuming the t distribution. Thus, we report the results of a Spearman rank correlation test instead, which, unlike Pearson's correlation, is a non-parametric test without assuming a linear relationship between the variables, nor a normal distribution of the variables (Spearman, 1904).

All the correlation models had regularizing priors. For the RT models and the ranked FPRreg probability correlation models, position parameter μ followed a $\mathcal{N}(150, 100)$ distribution and scale parameter σ followed a $\mathcal{N}_+(150, 100)$ distribution. The μ prior suggested that the center of either MoTR or eye-tracking data was likely within a range of 350. The wide σ prior reflected our uncertainty about the variability present in the two data sets. The ν parameter, governing the degrees of freedom for the student t -distributions, was modeled with a $\text{Gamma}(2, 0.1)$, making a wide range of values, from 0 to 80 to be very likely. The ρ parameter was uninformative, following $\text{Uniform}(-1, 1)$. For modeling skip and first pass association probability, the priors for μ and σ were narrowed to $\mathcal{N}(0, 1)$ and $\mathcal{N}_+(0, 1)$. We used four chains with 4000 iterations of which the first 2000 were warm-up. The convergence of the models was verified by examining the $\hat{R}$ -statistic and trace plots (Gelman et al., 1995). We fit models using Stan (Carpenter et al., 2017). When fitting models for the BSPPR, our models did not successfully converge, likely due to the extremely small number of skips and regressions in the data. In BSPPR, we observed a regression rate and

a skip rate of 1%, whereas in eye-tracking we observed a regression rate of 16% and a skip rate of 44% and in MoTR we observed a regression rate of 4% and a skip rate of 46%. Instead of reporting correlations from non-converged models we simply mark the BSPPR correlations for these metrics with a dash in tables.

We visualize the relationship between MoTR/BSPPR and eye-tracking data in Fig. 5. Here, the solid black line shows the $x = y$ line, and the dotted black line shows the linear line of best fit. The left panels show the relationship for the reading time measures, whereas the right-hand panels show the relationship for regression and first-pass association probabilities. Each point is the average across all participants for a given word. The largest differences between MoTR and BSPPR can be observed in the right-hand panels, where BSPPR data clusters around zero (for regression probability) or one (for first-pass association probability). The MoTR data displays more spread, especially for the association probabilities. Turning to the reading time measures, there are two high-level trends worth noting: First, both the MoTR and BSPPR data seem, at least visually, to be linearly correlated with the eye-tracking data. Second, for these measures, the MoTR and BSPPR data are rightward skewed, meaning that their reading times tend to be slightly longer than eye-tracking reading times for the same word.

How well do the MoTR and BSPPR data correlate with the eye-tracking data? To compute an upper bound, we split the eye-tracking data into two groups, based on the participant ID assigned in the Provo dataset. Participants with ID less than 42 were assigned to the first group, while those with ID greater than 42 were assigned to the second. To compute an upper bound for correlation, we simply compute the

Table 2

Correlation with word-level statistics: Posterior means and 95% credible intervals of the correlation coefficient of our Bayesian model.

	Log frequency	Length	Surprisal
MoTR	-0.58 [-0.60, -0.55]	0.68 [0.66, 0.70]	0.43 [0.40, 0.47]
BSPR	-0.30 [-0.34, -0.26]	0.31 [0.27, 0.35]	0.32 [0.28, 0.36]
Eye-Tracking	-0.60 [-0.63, -0.57]	0.64 [0.61, 0.66]	0.51 [0.48, 0.54]

correlation between the by-word average for each of the two groups, using the same set of models introduced before.

The results can be seen in Table 1, with the MoTR/eye-tracking correlation on the top row, the BSPR/eye-tracking correlations in the middle row, and the eye-tracking self correlations on the bottom. There are three high-level trends that we would like to highlight: First, for every reading time metric (i.e., gaze duration, go past time, and total duration), we find a higher correlation for MoTR than for BSPR, meaning that the MoTR RTs are more similar than BSPR RTs to the eye-tracking data. That being said, despite the overall robust correlations, neither metric is as well correlated with eye-tracking as eye-tracking is with itself, as might be expected. Second, the place where we observe the lowest correlations between MoTR and eye-tracking is for first pass regression probability, where the MoTR/eye-tracking correlation is 0.19 and the self eye-tracking correlation is 0.58. Third, we observe the highest correlation for skip probabilities, where the MoTR/eye-tracking correlation is very high, even approaching the levels observed for the eye-tracking self-correlations.

One potential concern is that while MoTR regression correlation is low (0.19), the correlation for go past time, which is largely determined by the presence or absence of regressions, is still relatively robust (0.42). How can this be the case? In Table 1 we find stronger correlations for gaze duration than for go past time. Therefore, adding information about regressions decreases the correlation between eye-tracking and MoTR, which is in line with what we would expect from the lower regression correlations. What this tells us is that the total number of regressions, both in eye-tracking and MoTR, is not large enough to impact the overall reasonably high correlation between the two metrics, when computed over a large dataset.

We hypothesize that the lower regression correlation in MoTR is due to the different amount of effort required to initiate a regression between the two paradigms. In eye-tracking, regressions (indeed, all saccades) are achieved by simply moving the gaze, however in MoTR participants must plan an execution of a more complicated hand movement, which must be coordinated with a gaze movement as well. One effect of this is that it could raise the activation barrier for regressive saccades. If we frame this in terms of trying to predict eye-tracking regressions from MoTR regressions, the high activation cost of the latter could result in a low number of *false positives*, but a high number of *false negatives*. That is, when a word triggers a regression in MoTR the same word is likely to trigger a regression in eye-tracking, but only a subset of the words that trigger eye-tracking regressions also trigger MoTR regressions. To test this hypothesis, we look at the distribution of regression-triggering words under each metric. Words were counted as regression-triggering words if any participant initiated a regression from it during its first pass. We found that 98% of regression-triggering words in MoTR were also regression-triggering in eye-tracking, however, only 38% of regression-triggering words from eye-tracking also triggered a regression in MoTR. This confirmed our hypothesis, that MoTR is good at picking up eye-tracking regressions with high precision, but only picks up a subset of the regressions that can be found in eye-tracking.

Relationship to word-level statistics

We investigate the relationship between MoTR, BSPR, eye-tracking, and various word-level statistical properties that have been shown to

impact reading times in naturalistic settings. Our word-level properties are log frequency, as reported in Speer (2022); word length in characters, and SURPRISAL. Surprisal is the negative log probability of a word, given its context, i.e., $s(w_i) = -\log p(w_i | \mathbf{w}_{<i})$, and is known to be a strong predictor of reading time above and beyond length and frequency (Hale, 2001; Levy, 2008; Smith & Levy, 2013; Wilcox et al., 2020, 2023). Here we look simply at the correlation, however, we conduct a more in-depth investigation into the functional relationship between surprisal and MoTR data in the next section. All of our surprisal values are estimated from GPT-2 (Radford et al., 2019), a large language model that has been trained to predict the distribution over sub-word units (called *tokens*) given their context. To turn predictions over tokens into predictions over words we simply sum the surprisal over within-word tokens, following the chain rule of probabilities. For this analysis, correlations are computed using the same methods as described in the previous section, however here we adjusted the μ prior to $\mathcal{N}(0, 100)$ and σ to $\mathcal{N}_+(0, 100)$. Note that our correlation analysis measures the relationship between the RT and statistical property on a single word, and therefore may be unable to capture certain structures in the data, such as local autocorrelated fluctuations in reading times, due to, for example, participants getting tired or impatient.

Fig. 6 shows the relationship between eye-tracking (left column), MoTR values (middle column), and BSPR values (right column) against our various word-level properties along the different rows. As the results are similar across the different metrics, we show only data for gaze duration. Correlations are reported in Table 2. We find similar trends as before: MoTR correlations are higher than BSPR correlations. Overall, we find that log frequency and length are more highly correlated with MoTR than surprisal. In the next section, we follow up on this observation and take a deeper look at the relationship between surprisal and MoTR reading times.

Analysis 3: The functional relationship between MoTR and surprisal

In this section, we investigate the functional relationship between MoTR RTs and surprisal. Following Smith and Levy (2013) and Wilcox et al. (2020), we produce a visualization of the relationship that can be inspected to determine if it is (roughly) linear. To do so, we use generalized additive models (GAMs), which are a class of models that can fit non-linear relationships between predictor variables and response variables. Given that the GAM has a less constrained hypothesis space than a linear model, if it finds a relationship that looks roughly linear, this is good evidence that the underlying effect is, itself, linear. In addition, following more recent work (Hoover et al., 2021; Meister et al., 2021; Shain et al., 2024; Wilcox et al., 2023), we complement these visualizations with statistical analyses that test for evidence for non-linearities in the relationship. To do so, we fit GAMs that enforce a linear relationship between surprisal and MoTR RTs. We then use leave-one-out cross-validation to assess whether these linear GAMs are worse at predicting held-out data than the non-linear GAMs. If we do not find a difference in predictive power between these classes of models, this suggests that the surprisal-RT linking function is best characterized as linear.

To conduct these analyses we fit GAMs to predict the reading time of a word w_i from its log frequency, length in characters, and surprisal as well as the log frequency, length, and surprisal of the previous word w_{i-1} . This was done due to the previous spillover effects which have been found for eye-tracking. As before, our frequencies are from Speer (2022), length is measured in the number of characters, and surprisal estimates are from GPT-2. In the GAM model, we include smooth terms for our surprisal values as well as tensor product terms for a non-linear interaction between log frequency and word length. GAMs are fit using the *mgcv* package in R.³ To capture non-linearities in this

³ A sample GAM call: `rt ~ s(surp(w_i), bs = 'cr', k = 6) + s(surp(w_{i-1}), bs = 'cr', k = 6) + t2(freq(w_i), len(w_i), bs = 'cr') + t2(freq(w_{i-1}), len(w_{i-1}), bs = 'cr')`.

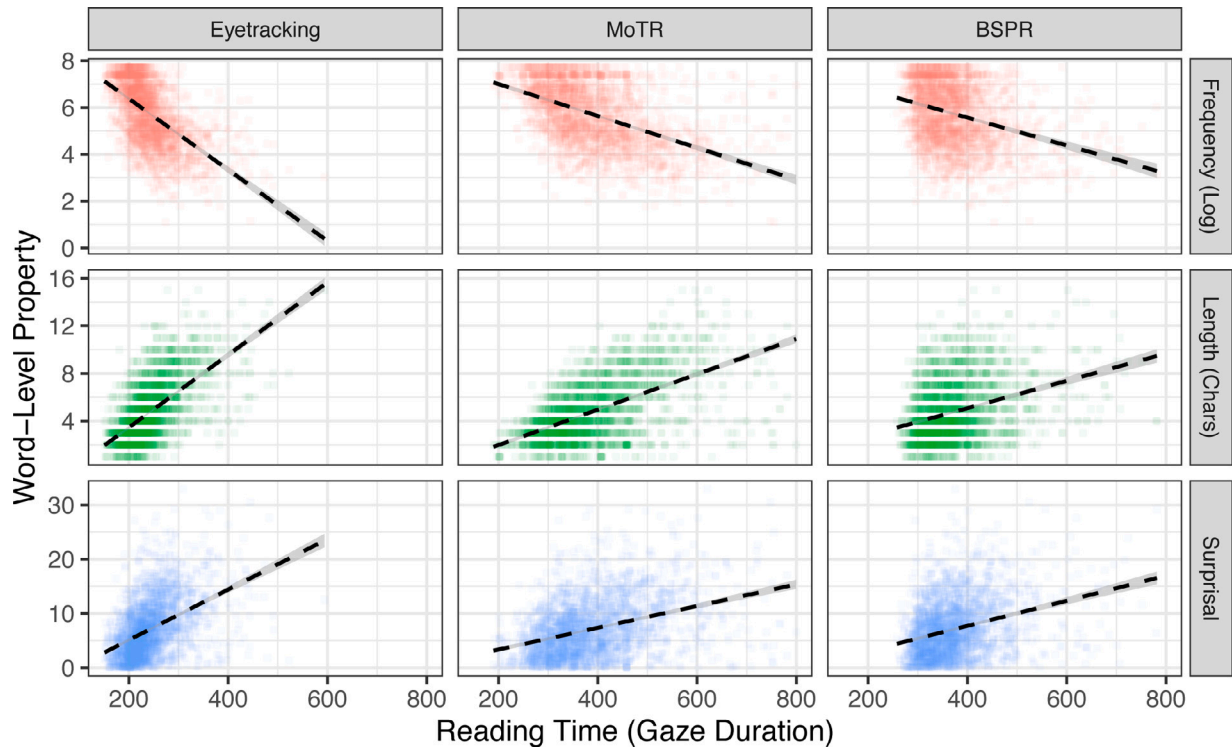


Fig. 6. Comparison between MoTR, BSPR, eye-tracking, and various word-level statistics: Results are for gaze duration.

model we set up a non-linear spline-based smooth for the effect of surprisal, which is restricted to six basis functions. The linear models are the same, except they include only linear terms for surprisal, instead of spline-based smooth terms. We train models on twenty different folds of data. For fitting the GAM model in a Bayesian framework, we assumed RTs follow a Gamma distribution with a log link function. The intercept was set to a normal distribution $\mathcal{N}(6, 1)$. The fixed effects were assigned a normal distribution prior $b \sim \mathcal{N}(0, 0.5)$, reflecting our uncertainty about the predictors' impact on RTs and allowing for potentially large effect sizes. For the same reason, the smooth terms followed a Cauchy distribution, $sds \sim \text{Cauchy}(0, 0.5)$, which was similar to normal distribution but with heavier tails, making it more tolerant of extreme values and less informative. The shape parameter followed an Exponential(2) distribution, providing flexibility while assuming an extremely long-tailed RT distribution to be unlikely. We used four chains each with 4000 iterations, of which the first 2000 were warm-ups. To ensure convergence, the fitting process is controlled with an `adapt_delta` parameter of 0.99.

To produce visualizations, we sample the models' predicted reading times based on the surprisal of w_i and w_{i-1} for values from 0–20 in increments of 0.1. These samples thus give us the models' predicted slowdown due to surprisal across a wide range of surprisal values. We can inspect if the predicted slowdown due to surprisal grows linearly as a function of surprisal itself. Samples from our twenty different models were used to compute 95% bootstrapped confidence intervals.

The result of this procedure can be seen in Fig. 7, with the non-linear fits in different colors, depending on the metric, and the linear fits behind them in gray. Slowdowns due to surprisal of the current word w_i are shown with solid lines, whereas slowdowns due to the surprisal of the previous word w_{i-1} are shown with dotted lines. We wish to draw out two points: First, we find a roughly linear relationship between surprisal and MoTR RTs. This is visually evident by the fact that samples from the linear and non-linear models are essentially superimposed on top of each other in Fig. 7. The second point is that effects for surprisal for w_i are about as large as those for w_{i-1} . This is different from what has been observed for strongly incremental processing measurements

such as the maze task, where the effect of surprisal of the previous word has been found to be essentially nonexistent (Boyce et al., 2020). However, it is in line with previous results for eye tracking, where the effect of surprisal has been found to be about equal for w_i and w_{i-1} (Smith & Levy, 2013). Overall, these data suggest that MoTR reading times do preserve the functional relationship found in eye tracking between processing times and surprisal, however we test this more rigorously below.

To test, statistically, whether there is evidence for a non-linear relationship between surprisal and MoTR RTs, we conduct a model comparison between our linear and non-linear GAMs for each of our reading measures. We used leave-one-out cross-validation to compare the predictive accuracy between the two types of models, which involves testing each model on a single observation while training it on the remainder of the dataset (Vehtari et al., 2015, 2017). This procedure produces a difference in log pointwise predictive density (Δ ELPD), where a negative value indicates higher predictive accuracy for the linear model. The findings align with the visual observations. We observe a consistently negative, albeit modest, Δ ELPD, regardless of the reading time measurement used. For gaze duration the Δ ELPD is -2.7 with a standard error of 0.9; for go past time it is -1.4 ($SE = 1.4$); for total duration it is -1.1 ($SE = 1.8$). These results indicate that, despite the increased parameter count of the non-linear models, the linear models are actually *better* at predicting MoTR reading times, likely because the non-linear models are overfitting to their training data. Overall, these data are in-line with the visualizations above and suggest that MoTR reading times do preserve the linear functional relationship found in eye-tracking between processing times and surprisal.

Experiment 2: Attachment preferences in English

In this experiment, we ask whether MoTR can measure well-studied processing patterns that have to do with attachment preference in English. Decades of psycholinguistic research has revealed that language processing is incremental, meaning that the processor builds syntactic representations on the fly and attempts to incorporate new material

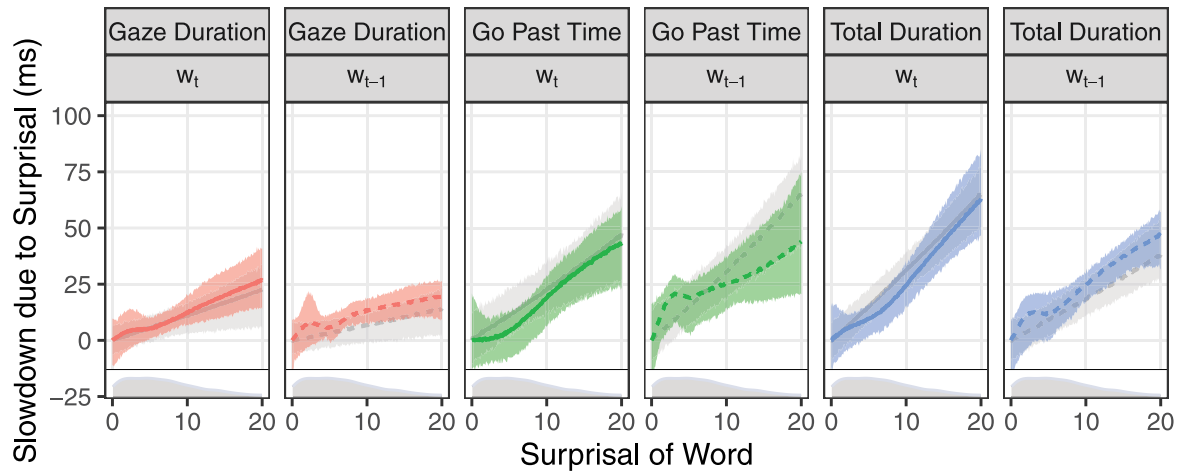


Fig. 7. Relationship between surprisal and MoTR reading times: Shaded regions are bootstrapped 95% confidence intervals. Solid lines indicate the slowdown due to the surprisal of the current word, w_t , whereas dotted lines indicate the slowdown due to surprisal of the previous word w_{t-1} . Gray lines show the relationship fit by GAMs that enforce a linear relationship between surprisal and MoTR RTs. Sub-plots at the bottom show the distribution of surprisal in the corpus.

into the representation as soon as it is encountered. Often, the grammar constrains how new syntactic material can be added to the incremental representations. However, sometimes the new material is syntactically ambiguous, insofar as it can be grammatically incorporated into the syntactic parse at multiple different locations.

When this is the case, where do people choose to incorporate the material? Frazier (1979) argues that native speakers of English often show a preference for incorporating the material into the phrase that is currently being processed (referred to as the LATE CLOSURE principle). This results in a processing strategy that favors lower attachment sites, i.e., ones that are further away from the root of the syntactic tree. However, subsequent experiments have demonstrated that the late closure principle is not operative in other languages (e.g., Spanish, Cueto & Mitchell, 1988), or even a general preference in English (Gilboy et al., 1995), and that other factors may influence attachment, such as modifying clause length (Hemforth et al., 2015) as well as expectations about discourse (Rohde et al., 2011). Despite this variability, preference for low attachment in English has become something of a testing ground for new incremental processing measures, having previously been used to study the difference between maze, SPR, and eye-tracking (Witzel et al., 2012), as well as between A-maze, L-maze, G-maze and self-paced reading (Boyce et al., 2020). In this section, we use the same materials as those originally presented in Witzel et al. (2012) (and also used by Boyce et al., 2020) to compare MoTR with these previous measurement tools.

Materials and methods

Materials

We test three different types of attachment preferences. Following Witzel et al. (2012) all of our materials are split into sentence regions, which are indicated with underlines in the examples below. Each sentence contains a pre-critical region, a critical region which is bolded in the examples below, and where we expect the strongest differences between conditions, as well as two post-critical regions, where we expect weaker effects. Region delineation is reconstructed from the examples given in Tables 1–6 of Witzel et al. (2012). In the next paragraphs, we walk through the materials for each of the three sub-experiments.

Adverb attachment. The first set of materials investigates the attachment of temporal adverbs, with the two conditions given in (1), below. Both conditions include a matrix clause with a verb (here, the past-tense *wrote*) as well as a relative clause with a verb whose tense

is mismatched to the main clause verb (here, the future tense *will deliver*). This is followed by an adverb whose temporal properties either match the relative clause or the matrix clause and force either a low attachment or high attachment reading. If low attachment is easier to process, then subjects should read this critical region more quickly when it agrees temporally with the relative clause, as opposed to the matrix clause.

(1) Adverb Attachment

- Dan wrote the speech he will deliver next month, but he hasn't practiced it yet. (LOW ATTACHMENT)
- Dan wrote the speech he will deliver last month, but he hasn't practice it yet. (HIGH ATTACHMENT)

Coordination attachment. The second set of materials test attachment preferences for coordination phrases, with conditions given in (2). Both conditions start with a transitive sentence and continue with a coordination phrase (*and the actor...* in the example below). In the NO COMMA condition, the phrase can plausibly be coordinated with the object of the transitive verb to form a larger constituent (i.e., *the actress and the actor*), however this structural interpretation is later ruled out by the presence of a later verb (*quit*, in the example below). If the reader has committed to this locally plausible but globally implausible parse, they should find the disambiguating verb unexpected, and we would expect relatively longer reading times on it in this condition. In the COMMA condition, a comma is inserted before the coordination phrase. In written text, commas can serve as cues to clause boundaries, rendering the coordination phrase unambiguously high-attached. We refer to this as the COORDINATION ATTACHMENT AMBIGUITY, or more simply as 'Coordination' in figures.

(2) Coordination Attachment

- The producer replaced the actress, and the actor quit the movie after the fight. (COMMA)
- The producer replaced the actress and the actor quit the movie after the fight. (NO-COMMA)

Relative clause attachment. The final set of materials tests preferences for relative clause attachment, with both conditions given in (3). Here, sentences start with a Noun Phrase that has an attached Prepositional Phrase. Both NP and PP contain a grammatically gendered animate noun, for example, *brother* and *ballerina* in (3). The sentence continues with a relative clause that includes a gendered reflexive pronoun. The pronoun either agrees with the gender of the second noun or the first

noun, forcing either high or low attachment for the whole relative clause. If participants are expecting low attachment, we should find faster reading times when the reflexive pronoun matches the second noun and the RC can be attached low.

(3) Relative Clause Attachment

- a. The brother of the ballerina who found **herself** in a lot of trouble phoned home (LOW ATTACHMENT)
- b. The brother of the ballerina who found **himself** in a lot of trouble phoned home (HIGH ATTACHMENT)

Methods

Participants were shown the same instruction screen as described in Section “Experiment 1: Naturalistic Reading in English” and were acclimatized to MoTR with three practice trials before moving on to the main experiment. Our materials were the same as those used in Witzel et al. (2012) and Boyce et al. (2020). They include 48 items for each attachment type, balanced between conditions, as well as twenty-four filler sentences. Comprehension questions were all binary yes/no questions, balanced equally between *yes* and *no* correct responses. Half of the comprehension questions were written for the original (Witzel et al., 2012) paper, while the other half were created later by Boyce et al. (2020). Questions from both batches follow the same format. The stimuli were partitioned into three lists using a Latin square design, such that no participant reads the same item in different conditions. Fillers were different between the lists, however practice items were not. In each experiment, practice items were shown, in the same order, to each participant, but fillers and target items were shuffled using built-in functions from Magpie. The font size was chosen so that materials would fit on a single line when tested on a 13-inch monitor.

We recruited 240 participants from Prolific; however, due to a submission error related to large data sizes in the experiment’s early phase – a problem that has since been resolved, as discussed in Section “Sampling Rate” – we successfully obtained data from 197 participants. Due to the large number of items in the experiment, we broke it into three sub-experiments with 59 items each, which took 14 minutes to complete, on average. After excluding 4 participants who scored less than 80% on comprehension questions, we were left with a total of 187 participants for analysis. In our setup, each of the participants reads 8 items per condition in each of the sub-experiments (adverb, coordination and relative clause attachment), meaning that we have $8 \times 187 = 1496$ observations per experimental condition. Participants were all native speakers of English, with IP addresses originating from within the United States. They were required to be using a desktop computer. While we cannot guarantee this, the choice of desktop was made so that participants would likely use a mouse and not a trackpad. Subjects were paid for their time and could not participate in more than one sub-experiment.

Data analysis

As in the first experiment, we used three criteria to discard participant data: First, data from a given participant was completely discarded if the participant got fewer than 80% of the comprehension questions correct. This threshold was chosen because it was the one used in Boyce et al. (2020). Ten participants were excluded for this reason. Additionally, data from individual trials was excluded if the participant fixated on fewer than 20% of the total words in the sentence, indicating that they were simply skimming. Words that were skipped during the first pass were not included in the computation of gaze durations and go past times. Finally, RTs were excluded if they were below f_{min} or above f_{max} , as described in Section “Mouse Tracking for Reading (MoTR)”.

For visualizations, we calculated the mean cross-participant per-word reading time for each sentence region and plotted the difference in means between conditions across items. For statistical tests, we fit a Bayesian hierarchical model to the mean within-region per-word reading time separately for each measure, or the mean FPrep probability for

each sentence region. The models included the experimental condition (HIGH ATTACHMENT VS. LOW ATTACHMENT OR COMMA VS. NO COMMA) as the sole fixed effect coded with sum contrasts (with HIGH ATTACHMENT being coded as 0.5 and LOW ATTACHMENT as -0.5), as well as by-item and by-subject random effects to capture the variability associated with both subjects and items and the correlation between the random effects. For our continuous reading measures (gaze duration, go past time, and total duration), we assumed a log-normal likelihood. For regressions, which were binary variables (i.e., whether or not the participant regressed), we used a Bernoulli distribution, and the relationship between the predictors and the response was modeled using a logit link function.

We used weakly informative priors in all our models to integrate domain expertise on RTs and regression probabilities from related real-time processing methodologies (e.g., eye-tracking), while also accommodating methodological differences. For models predicting continuous measures (i.e., RTs), the intercept was set to follow a normal distribution $\mathcal{N}(6, 1)$, translating to an expected grand mean RTs of $\exp(6) \approx 403$ ms and a 95% RTs range of $\exp(4) \approx 54$ ms to $\exp(8) \approx 2980$ ms. The slope’s prior was configured to follow a normal distribution $\mathcal{N}(0, 0.1)$, and variance components were modeled with Exponential(2) to speed up sampling (McElreath, 2018). These choices were informed by Schad et al. (2021) and Nicenboim et al. (2023), who analyzed related reading studies (i.e., Nicenboim et al., 2016; Vasishth et al., 2018) and our anticipation that the variability and differences in RTs across conditions would not be dramatic (i.e., over 800 ms). For the correlation matrix within the random effects variance-covariance structure, an LKJ prior was used with a shape parameter $\nu = 2$ to reduce the influence of extreme correlations of ± 1 .

In the FPrep models, the intercept prior was $\mathcal{N}(0, 2)$, signifying the unbiased starting point expectation in log-odds space. This translated to a baseline probability of 0.5 for initiating a regression, indicating no strong prior assumption either way. The slope and the variance components followed $\mathcal{N}(0, 1)$ and $\mathcal{N}_+(0, 1)$ priors, respectively, implying that we do not have a prior belief that different conditions or random effects will drastically increase or decrease the regression probability. We used the same LKJ prior as for the continuous dependent variables. These choices were inspired by and aligned with Mertzen et al. (2023).

Each model ran on four chains with 4000 iterations; the first 2000 served as warm-up and were excluded from inference. The convergence of the models was verified by examining the $\hat{R}$ -statistic and trace plots (Gelman et al., 1995). Reading time effects are reported in milliseconds, having been transformed back from the log-scale (Nicenboim et al., 2023). In the text, we report the mean and 95% credible intervals (CrIs) for the model’s posterior distribution of the effect of the experimental condition. If the 95% CrIs do not cross zero, we take this as good evidence for an effect of condition on participants’ reading times. All models were fit using the *brms* package (Bürkner, 2017) in R, which fits models using Stan (Carpenter et al., 2017).

Results

Main results

The results of our Bayesian models can be seen in Fig. 8, which shows the posterior mean and 95% credible intervals for the sole fixed effect of condition in our model. Although this figure shows modeling results for multiple measurements (MoTR, SPR, and maze), we will focus on MoTR results here and discuss the comparison between measurements in the next section. In addition to these modeling results, Fig. 9 shows the means of the raw RT data. To give a sense of the comparison between MoTR and eye-tracking, we plot the eye-tracking differences between the condition/region averages reported in Witzel et al. (2012) in teal alongside our own data.

Focusing first on the adverb experiment, we found that high attachment leads to longer reading times in the critical region for all metrics: posterior mean β -weight and 95% Bayesian credible interval for gaze duration $\beta = 35$ [20, 51]; for go past time $\beta = 80$ [50, 115];

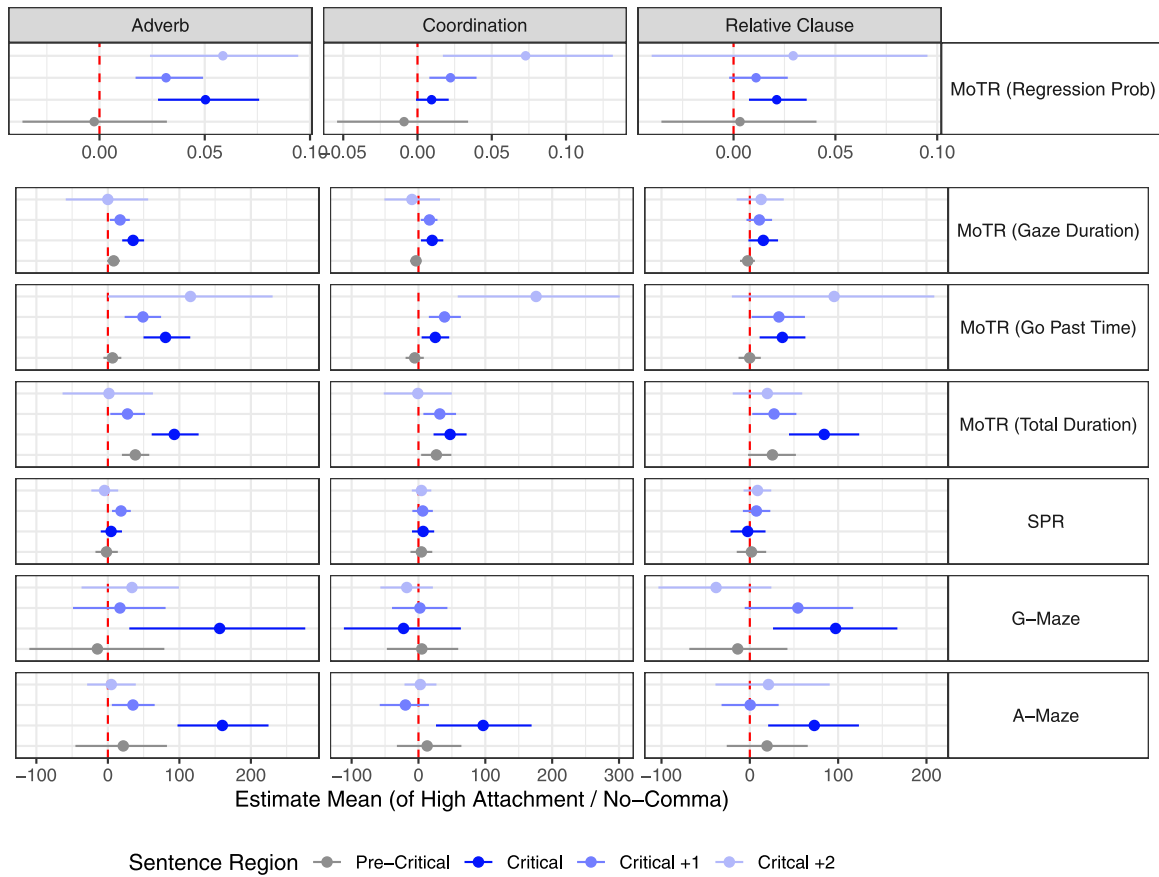


Fig. 8. Model estimates: Posterior means and 95% Credible Intervals for the main effect of experimental condition from our Bayesian logistic regression model (top panel) and linear regression models (bottom panels). The x-axis scale is in percentages of participants who regress for the top row, and in milliseconds of reading time for all other rows.

and for total duration $\beta = 93$ [61, 127]. In addition, we find that high attachment leads to longer reading times in the critical+1 region for all metrics as well (for gaze duration: $\beta = 17$ [3, 31]; for go past time $\beta = 49$ [23, 74]; and for total duration: $\beta = 27$ [4, 52]). We find that high attachment leads to increased total duration reading times in the pre-critical condition $\beta = 38$ [20, 58], indicating that participants are re-reading this region in the high-attachment condition and not in the LOW ATTACHMENT condition. Finally, we find that HIGH ATTACHMENT leads to greater instances of regressions in the critical region $\beta = 0.05$ [0.03, 0.08], critical +1 region $\beta = 0.03$ [0.02, 0.05], and the critical +2 region $\beta = 0.06$ [0.02, 0.09], which is in line with the relatively larger effects found in go past time over gaze duration, where time spent during the regression is factored into the former but not the latter.

In the coordination attachment experiment, we found that NO COMMA leads to longer reading times in the critical region for all metrics (for gaze duration: $\beta = 21$ [3, 37]; go past time: $\beta = 25$ [5, 46]; and total duration: $\beta = 47$ [23, 72]). We also find that the NO COMMA condition leads to longer reading times in the critical +1 region for all metrics as well (for gaze duration: $\beta = 16$ [4, 28]; for go past time: $\beta = 39$ [16, 63]; and for total duration $\beta = 32$ [7, 56]). In line with the adverb experiment, for total duration, we also find longer reading times in the pre-critical region in the NO COMMA condition $\beta = 27$ [4, 49], indicating that participants are making some regressions to this region in the NO COMMA condition. Unlike the Adverb experiment, we do not find an effect of condition on regression probability in the critical region. Rather, regression probability in the NO COMMA condition increases in the regions following the disambiguating material (for the critical +1 region: $\beta = 0.02$ [0.01, 0.04]; and for the critical +2 region: $\beta = 0.07$ [0.02, 0.13]).

For the relative clause experiment, we found that high attachment leads to reading time slowdowns in the critical region for go past time:

$\beta = 37$ [11, 63]; and for total duration: $\beta = 84$ [44, 124], as well as slowdowns in the critical +1 region (for go past time $\beta = 33$ [2, 62]; for total duration $\beta = 27$ [3, 53]). However, we did not find this effect for gaze duration. Additionally, we find that high attachment leads to increased regressions in the critical region $\beta = 0.02$ [0.01, 0.04].

There are three high-level patterns worth commenting on in the data. The first high-level pattern is that the MoTR data is able to reveal differences in processing between the conditions, with participants taking longer to read critical regions in HIGH ATTACHMENT/ NO COMMA conditions across the different metrics. Second, there is a consistent pattern in the metrics based on the inclusion of regression information, with gaze duration revealing the smallest differences between conditions, go past time revealing medium differences, and total duration revealing the largest differences difference between conditions, on the original millisecond scale. They suggest that, in a controlled psycholinguistic setting, MoTR can reveal how regressions out of and into sentence regions are implicated in sentence processing.

The final pattern worth mentioning is the tight connection we observe between MoTR and eye-tracking data in terms of the magnitudes of the effects. Although, as we saw in Section “Experiment 1: Naturalistic Reading in English”, participants take longer to read words in MoTR, on average, the differences in reading times between conditions are well-matched with the differences observed in the eye-tracking data. The major qualitative difference between the two is that: (a) effects at critical regions are larger in MoTR for the relative clause experiment, and (b) participants tend to regress more to the post-critical regions in eye-tracking, as evidenced by the higher teal dots in the critical +1 conditions for total duration in the adverb and coordination experiments. This latter difference is likely because

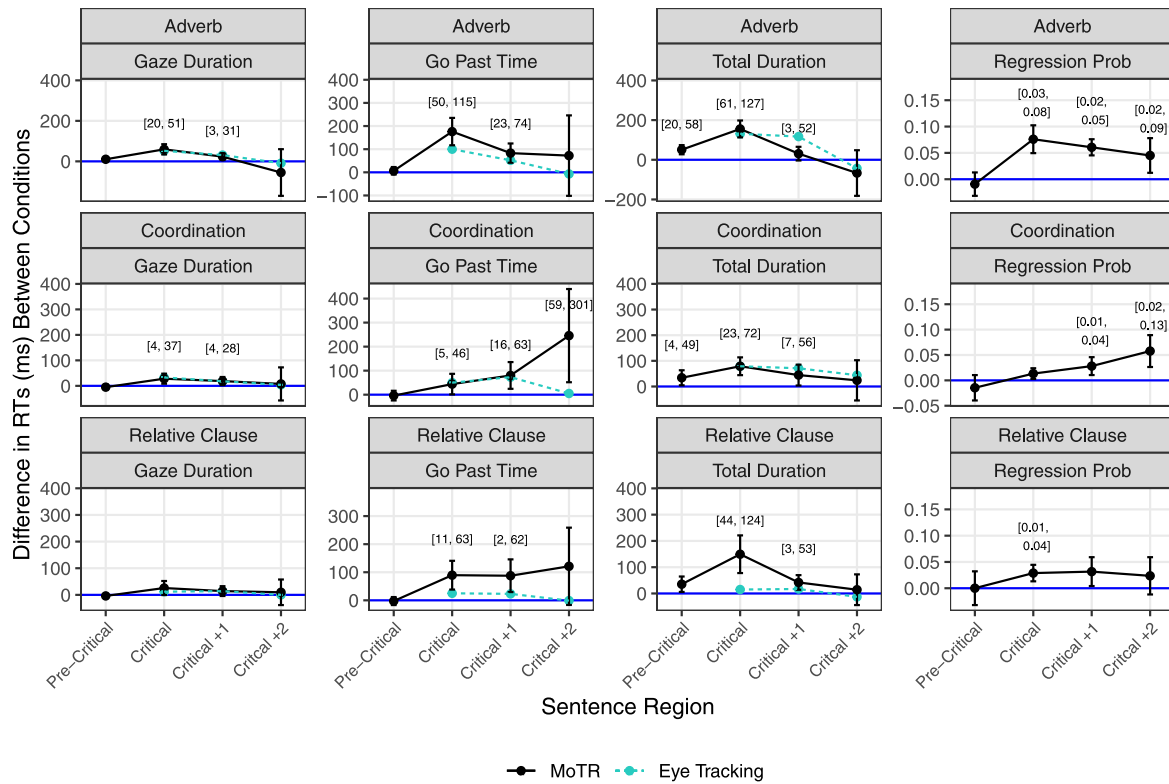


Fig. 9. Main results: Points are the by-item mean differences between conditions, where positive values indicate slowdown in HIGH ATTACHMENT/ NO COMMA conditions. The right column is the difference in first-pass regression probability, where the y-axis indicates the percentage of participants who regress. MoTR data is black, eye-tracking reported in Witzel et al. (2012) is teal. Error bars are 95% confidence intervals of the mean. The text indicates 95% credible intervals for the effect of condition from our models when these do not cross zero. The MoTR data reveals differences in processing difficulty between conditions and is well-matched to the eye-tracking data. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

regressions are less costly to initiate in eye-tracking, compared with MoTR.

Comparing MoTR to other web-based metrics

To get a sense of how MoTR compares to other web-based measures of incremental processing, we downloaded the data for web-based A-maze, G-maze, and SPR from Boyce et al. (2020) and analyzed it using our methods. We use the same methods as described above for filtering and fitting models on MoTR data, except for producing the visualizations, we also filter out RTs that are three standard deviations larger than the mean RT in each experiment. This was done due to the higher variance in the maze data. The results can be seen in Fig. 10. Unlike Boyce et al. (2020), who report word-by-word averages, we report average differences in reading times for the whole sentence region. We compare the maze and SPR data to MoTR go past time. Maze and SPR measure the amount of time that elapses from when the participant first sees a word, to when they progress past it in the sentence, which is essentially what go past time is measuring.

The four metrics differ somewhat in their ability to reveal differences in processing behavior between the conditions. Results for MoTR are the same as reported above: We find that the critical condition (i.e., HIGH ATTACHMENT OR NO COMMA) leads to slowdowns in all experiments. Using SPR, we find that high attachment leads to reading time slowdowns only in the adverb experiment critical +1 region $\beta = 18$ [6, 32]. Using G-maze, we find that high attachment leads to slowdowns in the critical region for the adverb experiment $\beta = 156$ [30, 276] and in the relative clause experiment $\beta = 97$ [26, 167]. Using A-maze, we find that high attachment leads to slowdowns in the adverb experiment $\beta = 160$ [97, 225] and the relative clause experiment $\beta = 73$ [21, 124], and that the NO COMMA condition leads to slowdowns in the coordination

experiment $\beta = 97$ [26, 169]. As far as effect sizes, A-maze and G-maze demonstrate the largest effect sizes, however, we also observe larger variance for these measurements, as indicated by the wider 95% credible intervals from our models. The tradeoff seems to be that maze will produce larger differences in reading times between conditions, but MoTR can pick up on how regressions are implicated in the processing of the phenomena.

Power analysis

The above results reveal that MoTR is capable of picking up differences between conditions in factorized experimental tests. In this section, we evaluate its sensitivity by conducting two types of power analyses, the first adapted from Jäger et al. (2020), Vasishth et al. (2018) and the second from Boyce et al. (2020). In both cases, we run our power analysis on MoTR go past times, because, as discussed above, this is the most sensible choice for making comparisons between MoTR, SPR, and maze.

Procedure 1: Varying effect size

The first power analysis asks how sensitive MoTR is at picking up differences between experimental conditions, assuming that such differences exist. To do so, we extracted parameter estimates, including both fixed effects estimates and variance components, from the Bayesian hierarchical model fits on MoTR data in Section “Data Analysis”. Following Jäger et al. (2020), we additionally selected three “counterfactual” sizes for our effect of interest (i.e., HIGH ATTACHMENT VS. LOW ATTACHMENT and COMMA VS. NO COMMA) based on the first quartile, median, and third quartile values derived from the posterior predictive distributions of the above models. We used the stored parameter estimates

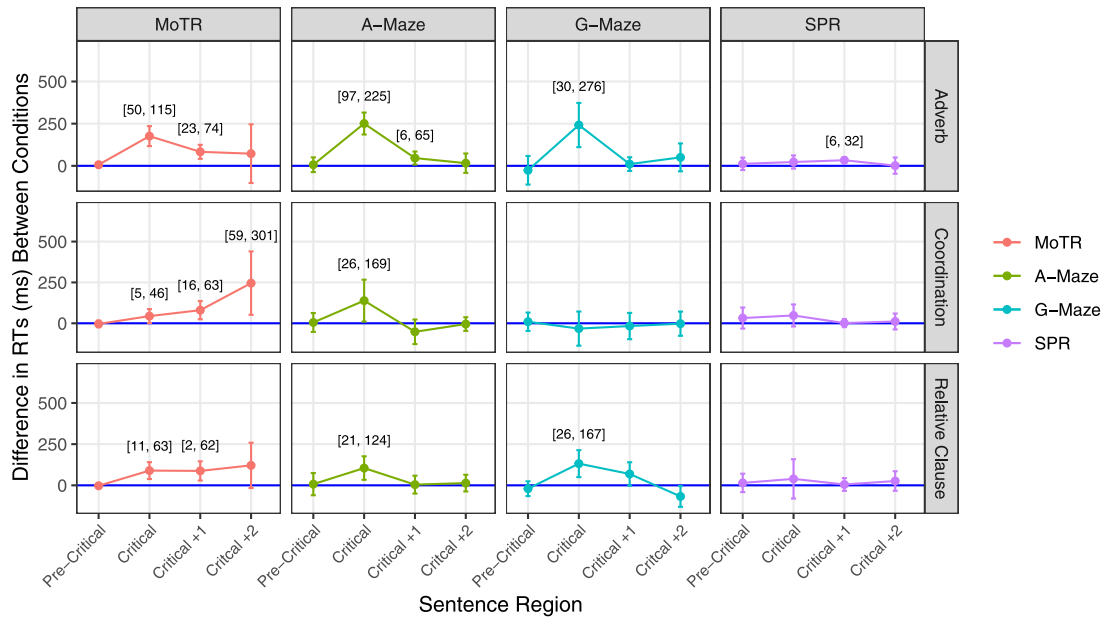


Fig. 10. Comparison between measures: Comparison between MoTR results (go past time) and data from three online incremental processing measurements reported in Boyce et al. (2020). Errors are 95% CIs across by-item differences; text indicates 95% credible intervals for the effect of condition from our regression models, when these do not cross zero.

Table 3

Power analysis, procedure 1: Results for three different “counterfactual” effect sizes. We use go past times and set the number of participants to 186 and assume that each participant produces 8 observations per condition, as was the case in our experiments.

Sub-Experiment	1st quartile	Median	3rd quartile
Adverb			
Effect magnitudes (ms)	69	80	91
Power estimates	0.99	1.00	1.00
Coordination			
Effect magnitudes (ms)	18	25	32
Power estimates	0.43	0.85	0.93
Relative Clause			
Effect magnitudes (ms)	28	37	45
Power estimates	0.71	0.89	0.97

and counterfactual effect sizes to generate 100 simulated datasets, with the same number of participants and items as in Experiment 2 (i.e., with 186 participants producing 8 observations per condition). Datasets were generated using `rlnorm()` function in R, where for each observation, we simulated a single random reading time value from a log-normal distribution. When simulating these data with `rlnorm()` we need to input two parameters – the mean and standard deviation of the distribution. To obtain the former, we calculate the posterior mean from our Bayesian model, taking into account the stored population-level parameter estimates (fixed effects), counterfactual effect sizes, and group-level effects (random effects). To obtain the variance, we take the family-specific variance parameter from the fitted Bayesian model. We simulate these datasets with the `rlnorm()` function as opposed to say, the models’ built-in simulate function because it is more straightforward to vary the effect sizes using this method. Once these simulated datasets have been constructed, we fit `lme4` models (Bates et al., 2015) – which were chosen for increased speed – on them and calculate the power as the proportion of these datasets for which the relevant predictor is statistically significant ($p < 0.05$).

We show the results from this procedure in Table 3. We find that the power is high for all the experiments – above 0.80 for the assumed true effect size across the board, and even above 0.4 for the lower quartile in all three cases. These results indicate that MoTR is an effective tool for

picking up differences between experimental conditions, assuming that these differences exist and that the participant count and observations per participant are similar to the ones in this study.

Procedure 2: Varying participant number

Here, we conduct a prospective power analysis for multiple different measurement modalities (maze, MoTR, and SPR) while varying the number of participants. This allows us to compare the sensitivity of MoTR to other methods. As opposed to the previous analysis, where we simulated counterfactual datasets with the `rlnorm()` function, here, we directly use the simulation function associated with the fitted Bayesian model. We generated 100 datasets for each modality, keeping the critical effect size constant, but now varying the number of simulated participants, from between 10 to 200. This analysis assumes that each participant produces 12 observations per experimental condition. We then calculate the power as the proportion of datasets for which the relevant predictor is statistically significant ($p < 0.05$). Power estimates are calculated for the critical sentence region.

Results are shown in Fig. 11. We find that, given our budget of 12 observations per experimental condition, about 40 participants are needed for an 80% power in the adverb attachment experiment, and 140 in the relative clause and coordination experiments. Overall, the sensitivity of MoTR lies inbetween SPR and maze, although note that MoTR is more sensitive than G-maze for the coordination experiment.

One question raised by our results is why G-maze has such low power in the Coordination experiment. We note that this lower power is also observed by Boyce et al. (2020), and consistent with the null result for G-maze at measuring differences in reading times between conditions in this experiment (as is visually evident in Fig. 10). We hypothesize that this lower power may be caused by the distractors in G-maze. Because of the coordinating conjunction, it is difficult to write distractors that are truly ungrammatical, and we found several examples of distractors that, in fact, do license plausible, grammatical continuations in the original materials.⁴

⁴ For example, item number 42 is The police arrested the burglar and . . . with a grammatical continuation of his and an ungrammatical

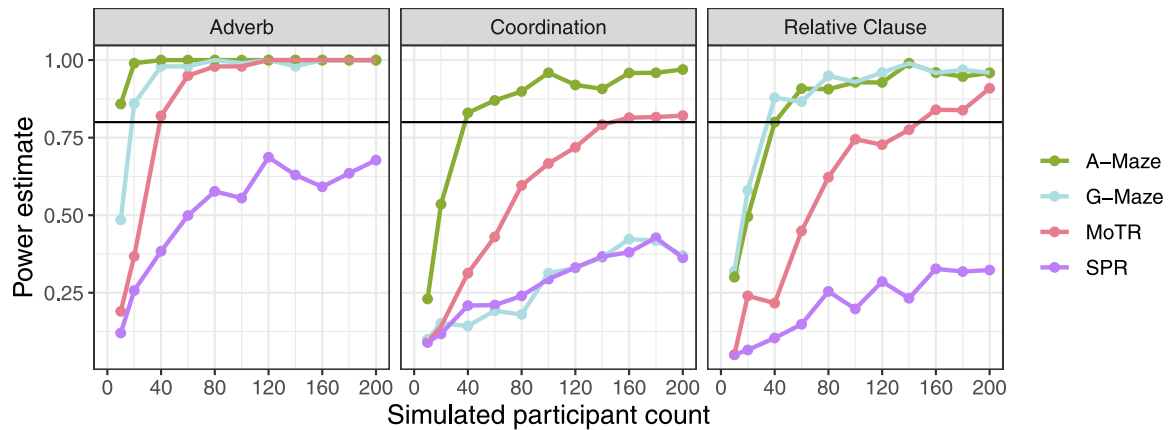


Fig. 11. Comparison of power between MoTR, SPR and maze: The power estimates were calculated on the critical sentence region. Comparisons are between MoTR go past time and data from the three incremental processing measurements reported in Boyce et al. (2020).

In sum, these results indicate that the number of participants needed for a MoTR experiment is within the range typically recruited for online psycholinguistic experiments. They validate one of the key premises of the method, namely that it can be used to collect psycholinguistic data on modest budgets.

Discussion

To demonstrate the potential utility of a MoTR experiment, we briefly discuss how it can be used to reveal processing differences between our three sub-experiments, something which is not evident in the maze data, nor obvious in the eye-tracking data. These differences have to do with the role of regressions during processing, specifically, when a regression is initiated. In the relative clause experiment, we observe that the difference in regression rate between the conditions falls only in the critical region itself. That is, participants are more likely to regress on the reflexive pronoun that forces high-attachment, but not subsequently in the sentence. For the adverb experiment, we observe a similar difference in regression rate in the critical region, however, this difference is sustained in both post-critical regions as well. This can be observed both in the higher difference in regression rates for these regions, as well as in the larger difference in go past time and total duration RTs observed in the critical, critical +1, and critical +2 regions for this experiment. In the coordination experiment, the pattern of results is quite different. Here, we observe no difference in regression rate between conditions in the critical region, but a steady increase of regression rate difference in the NO COMMA condition for both post-critical regions. This indicates that, although participants slow down at the disambiguating verb, they save their regressions for the end of the sentence.

At a high level, these patterns are consistent with two different processing strategies – an immediate regression strategy that is used during the processing of adverbs and relative clauses, and a sentence-final regression strategy that is used when processing sentences with coordination ambiguities. We would like to stress, however, that further experimental validation of these results should be conducted before strong theoretical conclusions are drawn.

continuation of why. However, it is possible to create grammatical sentences with this continuation; for example, The police arrested the burglar and why they did not prosecute him is a mystery to me. This example actually highlights one advantage of A-maze over G-maze, which is that it makes no promises about the true ‘ungrammaticality’ of distractors, merely that they will be relatively improbable under the distribution of some language model.

Turning back to the main question posed at the beginning of this section, we find that MoTR can, indeed, measure well-studied language processing patterns in English. Across experiments and metrics, we observe a robust slowdown in processing high-attachment sentences, which is consistent with prior results for these materials. Thus, the results from this experiment demonstrate that MoTR can be used to reveal processing behavior in a targeted psycholinguistic setting.

General discussion

We begin our discussion by turning back to the three considerations mentioned in the introduction – precision, naturalness, and cost. Starting, first, with cost, it is evident the MoTR is as cheap to run as the cheapest current alternatives, maze and SPR. Because it runs in the browser, it is possible to use the paradigm for studies that recruit large numbers of participants over the internet. Although we did not test this directly in our experiments, it is also possible to run MoTR experiments locally in the browser, meaning that MoTR can be used wherever computers are available, such as in labs not equipped with eye-trackers and, potentially, in fieldwork. Furthermore, our power analyses suggest that psycholinguistic processing patterns, such as attachment preferences, can be measured using MoTR with subject pools, of around 100 participants.

Turning to naturalness, we argue that this aspect of MoTR sets it apart from the other web-based measures, SPR and maze. In both traditional (moving window) SPR and maze participants are both unable to skip words and regress to previous words, making these techniques much more constrained than eye-tracking and potentially less ecologically valid as measurements of reading behavior. While making saccades in a MoTR trial requires coordinating hand and eye movement, our data suggests this does not inhibit participants from engaging in more naturalistic behavior that involves more than just a linear pass over the text. For example, in the Provo experiment, we find that, on average, participants make regressions in 4% of words and skip about 46% of words, compared to a regression rate of 15% and a skip rate of 44% in eye-tracking. While regression rates are certainly lower in MoTR, it is also clear that participants take advantage of the paradigm to make flexible scanpaths through the materials.

Turning to the question of precision, it is clear that eye-tracking still offers the most precise behavior measurements for psycholinguistic reading research. However, MoTR may have an advantage over eye-tracking when it comes to preview effects. Because it is difficult in eye-tracking studies to control the extent of parafoveal processing (unless in gaze contingency or boundary paradigms), the extent to which preview effects influence observed behavior is difficult to determine when it comes to eye-tracking. Because upcoming words are occluded

in MoTR (as well as in maze and SPR), this potential preview effect can be ruled out therefore excluding any effects of parafoveal processing that were not intended by the researcher or the experimental design. Unlike in maze and SPR, however, upcoming words in MoTR are not fully occluded (they are simply blurred) and the participant can partially reveal some upcoming material with the gradual transition effect of the spotlight. That being said, a MoTR experiment does enable the experimenter to know for certainty what was in focus and what was blurred and thus unreadable at any given point in time.

As far as precision, how does MoTR compare against SPR and maze? While SPR does not pick up on any of the differences between conditions in experiment two, both MoTR and A-maze show robust slowdowns in the HIGH ATTACHMENT and NO COMMA conditions. A-maze recovers larger magnitudes in processing times between conditions and is potentially more incremental insofar as most of its effects are solely on the critical region. In MoTR, some of the effects are in post-critical regions. However, MoTR still picks up robust differences between conditions and, to its advantage, its effect sizes are more in line with those observed in the eye-tracking data. We believe that MoTR and maze can offer useful complements to each other. In cases where only incremental processing slowdowns are necessary for evaluating theoretical claims, then maze may be the more revealing experimental tool. However, in cases where information about regression and skipping as well as processing slowdown are important, then MoTR may be the better choice.

We designed the experiments in this paper to offer a balance between the type of experiments that are traditionally used to assess psycholinguistic theories, including both targeted psycholinguistic minimal pairs and naturalistic materials. Furthermore, we have done so with materials for which previously collected maze, SPR, and eye-tracking data are already available. There is one major gap in our assessment, however, and this is that our studies were collected entirely in English. Testing MoTR on a more diverse range of languages and scripts other than the Latin script is an obvious next step in order to fully validate this new measurement.

CRedit authorship contribution statement

Ethan Gotlieb Wilcox: Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Cui Ding:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation. **Mrinmaya Sachan:** Supervision, Resources, Investigation, Funding acquisition. **Lena Ann Jäger:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The raw data used in our experiments can be found at <https://osf.io/4g7pr>. Analysis scripts, as well as code to run a MoTR experiment in the browser, can be found at <https://github.com/wilcoxeg/MoTR>.

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